# RAJALAKSHMI ENGINEERING COLLEGE

## RAJALAKSHMI NAGAR, THANDALAM – 602 105

## Laboratory Record

Name : PRAKASH A

Year / Branch / Section : III/ Computer Science And Design / A

Register No. : 251701041

Semester : III

Academic Year : 2026-2027

# RAJALAKSHMI ENGINEERING COLLEGE

# THANDALAM – 602 105.

## BONAFIDE CERTIFICATE

NAME PRAKASH A

ACADEMIC YEAR 2026-2027 SEMESTER III BRANCH CSD

UNIVERSITY REGISTER NO. 2116251701041

**Certified that this is the bonafide work done by the above student in the CD23321 -Python Programming for Design Laboratory during the year 2026 – 2027.**

**Signature of Faculty – in – Charge**

**Submitted for the Practical Examination held on .....**

**Internal Examiner** **External Examiner**

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Reg. No. : 2116251701041 Name : PRAKASH A

Year : 2026-2027 Branch : CSD Sec : A

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# CO23321-Python Programming for Design-2025 Batch

| | |
|---|---|
| Started on | Thursday, 23 July 2026, 4:26 PM |
| State | Finished |
| Completed on | Friday, 24 July 2026, 2:44 PM |
| Time taken | 22 hours 17 mins |
| Grade | 100.00 out of 100.00 |

Question 1 Correct Mark 10.00 out of 10.00 Flag question

Write a program to print the sum of even numbers and the sum of odd numbers in a given range.

Note: The sign of the input numbers should be ignored.

Input format:

First input number is an int type. If the 1st one of the two digits should be '1'.

If the input numbers are 247 and -154, the 1st one of the two digits should be '1'.

If the input numbers are 247 and -154, the 2nd one of the two digits should be '2'.

If the input numbers are -247 and -154, the 2nd one of the two digits should be '2'.

For example:

| Input | Result |
|---|---|
| 247<br>-154 | 1<br>2 |
| 247<br>154 | 2<br>2 |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
a=abs(a)
b=abs(b)
print(a+b)
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 247<br>-154 | 1<br>2 | 1<br>2 |
| ✓ | 247<br>154 | 2<br>2 | 2<br>2 |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 2 Correct Mark 10.00 out of 10.00 Flag question

Write a program to find the sum of the digits of the given numbers.

Input format:

Input consists of two integers that corresponds to the age and weight of a person respectively.

Output format:

Display "Legal to donate.

Display "Out of no eligible.

Sample Input

```
45
70
```

Sample Output

```
Yes
```

For example:

| Input | Result |
|---|---|
| 45<br>70 | Yes |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
if a>=45 and b>=50:
    print("Yes")
else:
    print("No")
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 45<br>70 | Yes | Yes |
| ✓ | 45<br>50 | Yes | Yes |
| ✓ | 45<br>49 | No | No |
| ✓ | 44<br>50 | No | No |
| ✓ | 44<br>49 | No | No |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 3 Correct Mark 10.00 out of 10.00 Flag question

Write a program to print the sum of even numbers and the sum of odd numbers in a given range.

Note: The sign of the input numbers should be ignored.

Input format:

First input number is an int type. If the 1st one of the two digits should be '1'.

If the input numbers are 247 and -154, the 1st one of the two digits should be '1'.

If the input numbers are 247 and -154, the 2nd one of the two digits should be '2'.

If the input numbers are -247 and -154, the 2nd one of the two digits should be '2'.

For example:

| Input | Result |
|---|---|
| 247<br>-154 | 1<br>2 |
| 247<br>154 | 2<br>2 |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
a=abs(a)
b=abs(b)
print(a+b)
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 247<br>-154 | 1<br>2 | 1<br>2 |
| ✓ | 247<br>154 | 2<br>2 | 2<br>2 |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 4 Correct Mark 10.00 out of 10.00 Flag question

Write a program to find the sum of the digits of the given numbers.

Input format:

Input consists of two integers that corresponds to the age and weight of a person respectively.

Output format:

Display "Legal to donate.

Display "Out of no eligible.

Sample Input

```
45
70
```

Sample Output

```
Yes
```

For example:

| Input | Result |
|---|---|
| 45<br>70 | Yes |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
if a>=45 and b>=50:
    print("Yes")
else:
    print("No")
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 45<br>70 | Yes | Yes |
| ✓ | 45<br>50 | Yes | Yes |
| ✓ | 45<br>49 | No | No |
| ✓ | 44<br>50 | No | No |
| ✓ | 44<br>49 | No | No |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 5 Correct Mark 10.00 out of 10.00 Flag question

Write a program to print the sum of even numbers and the sum of odd numbers in a given range.

Note: The sign of the input numbers should be ignored.

Input format:

First input number is an int type. If the 1st one of the two digits should be '1'.

If the input numbers are 247 and -154, the 1st one of the two digits should be '1'.

If the input numbers are 247 and -154, the 2nd one of the two digits should be '2'.

If the input numbers are -247 and -154, the 2nd one of the two digits should be '2'.

For example:

| Input | Result |
|---|---|
| 247<br>-154 | 1<br>2 |
| 247<br>154 | 2<br>2 |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
a=abs(a)
b=abs(b)
print(a+b)
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 247<br>-154 | 1<br>2 | 1<br>2 |
| ✓ | 247<br>154 | 2<br>2 | 2<br>2 |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 6 Correct Mark 10.00 out of 10.00 Flag question

Write a program to find the sum of the digits of the given numbers.

Input format:

Input consists of two integers that corresponds to the age and weight of a person respectively.

Output format:

Display "Legal to donate.

Display "Out of no eligible.

Sample Input

```
45
70
```

Sample Output

```
Yes
```

For example:

| Input | Result |
|---|---|
| 45<br>70 | Yes |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
if a>=45 and b>=50:
    print("Yes")
else:
    print("No")
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 45<br>70 | Yes | Yes |
| ✓ | 45<br>50 | Yes | Yes |
| ✓ | 45<br>49 | No | No |
| ✓ | 44<br>50 | No | No |
| ✓ | 44<br>49 | No | No |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 7 Correct Mark 10.00 out of 10.00 Flag question

Write a program to print the sum of even numbers and the sum of odd numbers in a given range.

Note: The sign of the input numbers should be ignored.

Input format:

First input number is an int type. If the 1st one of the two digits should be '1'.

If the input numbers are 247 and -154, the 1st one of the two digits should be '1'.

If the input numbers are 247 and -154, the 2nd one of the two digits should be '2'.

If the input numbers are -247 and -154, the 2nd one of the two digits should be '2'.

For example:

| Input | Result |
|---|---|
| 247<br>-154 | 1<br>2 |
| 247<br>154 | 2<br>2 |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
a=abs(a)
b=abs(b)
print(a+b)
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 247<br>-154 | 1<br>2 | 1<br>2 |
| ✓ | 247<br>154 | 2<br>2 | 2<br>2 |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 8 Correct Mark 10.00 out of 10.00 Flag question

Write a program to find the sum of the digits of the given numbers.

Input format:

Input consists of two integers that corresponds to the age and weight of a person respectively.

Output format:

Display "Legal to donate.

Display "Out of no eligible.

Sample Input

```
45
70
```

Sample Output

```
Yes
```

For example:

| Input | Result |
|---|---|
| 45<br>70 | Yes |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
if a>=45 and b>=50:
    print("Yes")
else:
    print("No")
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 45<br>70 | Yes | Yes |
| ✓ | 45<br>50 | Yes | Yes |
| ✓ | 45<br>49 | No | No |
| ✓ | 44<br>50 | No | No |
| ✓ | 44<br>49 | No | No |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 9 Correct Mark 10.00 out of 10.00 Flag question

Write a program to print the sum of even numbers and the sum of odd numbers in a given range.

Note: The sign of the input numbers should be ignored.

Input format:

First input number is an int type. If the 1st one of the two digits should be '1'.

If the input numbers are 247 and -154, the 1st one of the two digits should be '1'.

If the input numbers are 247 and -154, the 2nd one of the two digits should be '2'.

If the input numbers are -247 and -154, the 2nd one of the two digits should be '2'.

For example:

| Input | Result |
|---|---|
| 247<br>-154 | 1<br>2 |
| 247<br>154 | 2<br>2 |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
a=abs(a)
b=abs(b)
print(a+b)
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 247<br>-154 | 1<br>2 | 1<br>2 |
| ✓ | 247<br>154 | 2<br>2 | 2<br>2 |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 10 Correct Mark 10.00 out of 10.00 Flag question

Write a program to find the sum of the digits of the given numbers.

Input format:

Input consists of two integers that corresponds to the age and weight of a person respectively.

Output format:

Display "Legal to donate.

Display "Out of no eligible.

Sample Input

```
45
70
```

Sample Output

```
Yes
```

For example:

| Input | Result |
|---|---|
| 45<br>70 | Yes |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
if a>=45 and b>=50:
    print("Yes")
else:
    print("No")
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 45<br>70 | Yes | Yes |
| ✓ | 45<br>50 | Yes | Yes |
| ✓ | 45<br>49 | No | No |
| ✓ | 44<br>50 | No | No |
| ✓ | 44<br>49 | No | No |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 11 Correct Mark 10.00 out of 10.00 Flag question

Write a program to find the sum of the digits of the given numbers.

Input format:

Input consists of two integers that corresponds to the age and weight of a person respectively.

Output format:

Display "Legal to donate.

Display "Out of no eligible.

Sample Input

```
45
70
```

Sample Output

```
Yes
```

For example:

| Input | Result |
|---|---|
| 45<br>70 | Yes |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
if a>=45 and b>=50:
    print("Yes")
else:
    print("No")
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 45<br>70 | Yes | Yes |
| ✓ | 45<br>50 | Yes | Yes |
| ✓ | 45<br>49 | No | No |
| ✓ | 44<br>50 | No | No |
| ✓ | 44<br>49 | No | No |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 12 Correct Mark 10.00 out of 10.00 Flag question

Write a program to find the sum of the digits of the given numbers.

Input format:

Input consists of two integers that corresponds to the age and weight of a person respectively.

Output format:

Display "Legal to donate.

Display "Out of no eligible.

Sample Input

```
45
70
```

Sample Output

```
Yes
```

For example:

| Input | Result |
|---|---|
| 45<br>70 | Yes |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
if a>=45 and b>=50:
    print("Yes")
else:
    print("No")
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 45<br>70 | Yes | Yes |
| ✓ | 45<br>50 | Yes | Yes |
| ✓ | 45<br>49 | No | No |
| ✓ | 44<br>50 | No | No |
| ✓ | 44<br>49 | No | No |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 13 Correct Mark 10.00 out of 10.00 Flag question

Write a program to find the sum of the digits of the given numbers.

Input format:

Input consists of two integers that corresponds to the age and weight of a person respectively.

Output format:

Display "Legal to donate.

Display "Out of no eligible.

Sample Input

```
45
70
```

Sample Output

```
Yes
```

For example:

| Input | Result |
|---|---|
| 45<br>70 | Yes |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
if a>=45 and b>=50:
    print("Yes")
else:
    print("No")
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 45<br>70 | Yes | Yes |
| ✓ | 45<br>50 | Yes | Yes |
| ✓ | 45<br>49 | No | No |
| ✓ | 44<br>50 | No | No |
| ✓ | 44<br>49 | No | No |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Question 14 Correct Mark 10.00 out of 10.00 Flag question

Write a program to find the sum of the digits of the given numbers.

Input format:

Input consists of two integers that corresponds to the age and weight of a person respectively.

Output format:

Display "Legal to donate.

Display "Out of no eligible.

Sample Input

```
45
70
```

Sample Output

```
Yes
```

For example:

| Input | Result |
|---|---|
| 45<br>70 | Yes |

Answer: (penalty regime: 0 %)

```
#Enter your code here
a=int(input())
b=int(input())
if a>=45 and b>=50:
    print("Yes")
else:
    print("No")
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | 45<br>70 | Yes | Yes |
| ✓ | 45<br>50 | Yes | Yes |
| ✓ | 45<br>49 | No | No |
| ✓ | 44<br>50 | No | No |
| ✓ | 44<br>49 | No | No |

Passed all tests! ✓

Marks for this submission: 10.00/10.00.

Finish review

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Finish review

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# CS23321-Python Programming for Design-2025 Batch

**Started on** Friday, 24 July 2026, 2:46 PM
**State** Finished
**Completed on** Thursday, 30 July 2026, 7:52 PM
**Time taken** 5 days 5 hours
**Marks** 10.00/10.00
**Grade** 100.00 out of 100.00

Quiz navigation
1 2 3 4 5 6 7 8 9 10
Finish review

**Question 1** Correct - Mark 1.00 out of 1.00 Flag question

A is assigned the first letter in the Alphabet B is assigned the second letter, and so on. Write a program that prints a list of the letters of the alphabet like the following in Scratch: you are formally to print the list of printable alphabet in each location.

Input:
Type: int
Length
Output:
Printable Characters

Input Format:
First Line contains the length of the list
Output Format:
All Letters of printable
All Letters of printable

**For example:**

| Input | Result |
|---|---|
| 26 | Total Number is: 26<br>All Letters of printable<br>ABCDEFGHIJKLMNOPQRSTUVWXYZ |
| 10 | Total Number is: 10<br>All Letters of printable<br>ABCDEFGHIJ |

**Answer:** (penalty regime: 0 %)

```
n = int(input())
print("Total Number is:",n)
print("All Letters of printable")
for i in range(n):
    print(chr(65+i),end="")
```

Reset answer

Test | Input | Result
1 | 26 | Total Number is: 26<br>All Letters of printable<br>ABCDEFGHIJKLMNOPQRSTUVWXYZ
2 | 10 | Total Number is: 10<br>All Letters of printable<br>ABCDEFGHIJ

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 2** Correct - Mark 1.00 out of 1.00 Flag question

Write a program to convert integer to an integer and float and display to type.

Sample Input:
5
6.5
Sample Output:
<class 'int'>
<class 'float'>

**For example:**

| Input | Result |
|---|---|
| 10 | <class 'int'><br><class 'float'> |

**Answer:** (penalty regime: 0 %)

```
a=int(input())
print(type(a))
print(type(float(a)))
```

Reset answer

Test | Input | Result
1 | 10 | <class 'int'><br><class 'float'>
2 | 20.5 | <class 'int'><br><class 'float'>
3 | 5 | <class 'int'><br><class 'float'>
4 | 7.5 | <class 'int'><br><class 'float'>
5 | 1000 | <class 'int'><br><class 'float'>
6 | 1000.5 | <class 'int'><br><class 'float'>

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 3** Correct - Mark 1.00 out of 1.00 Flag question

Write a program that returns the last digit of the given number. Last digit is being referred to the least significant digit i.e. the digit in the one's place in the number.

The last digit should be returned as a positive number.

For example,

If the given number is 432, the last digit is 2

If the given number is -107, the last digit is 7

**For example:**

| Input | Result |
|---|---|
| 432 | 2 |
| -107 | 7 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
print(abs(n)%10)
```

Reset answer

Test | Input | Result
1 | 432 | 2
2 | -107 | 7

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 4** Correct - Mark 1.00 out of 1.00 Flag question

Jumbo is a carpenter who works in Scratch. He decides it is completely time for a jumbo for him as there are weekdays and he fits for at least six weekends. He wants to find out how many days he has to work before that. If the value is 0 or 7 or more, he will not work at all. Help him to find the number of days he has worked on weekdays and weekends.

Input:
Type: int
Length of the number is 1, as we need that is 1st using data function
The day function returns the absolute value of the given number.

Output:
Weekdays
Weekends

**Sample Output:**
If the input is 0 or 7 or more
Jumbo will not work at all

**For example:**

| Input | Result |
|---|---|
| 0 | Jumbo will not work at all |
| 7 | Jumbo will not work at all |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
if(n==0 or n>=7):
    print("Jumbo will not work at all")
```

Reset answer

Test | Input | Result
1 | 0 | Jumbo will not work at all
2 | 7 | Jumbo will not work at all

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 5** Correct - Mark 1.00 out of 1.00 Flag question

Write a program that returns the second last digit of the given number. Second last digit is being referred the digit in the ten's place in the given number.

For single digit numbers, the second last digit is 0.

Note1 : The second last digit should be returned as a positive number, i.e. if the given number is -717, the second last digit is 1.

Note2 : If the given number is 0, then also the second last digit is 0.

For example,

If the given number is 432, the second last digit is 3

If the given number is -107, the second last digit is 0

**For example:**

| Input | Result |
|---|---|
| 432 | 3 |
| -107 | 0 |
| 0 | 0 |
| -10 | 1 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
print(abs(n)//10%10)
```

Reset answer

Test | Input | Result
1 | 432 | 3
2 | -107 | 0
3 | 0 | 0
4 | -10 | 1

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 6** Correct - Mark 1.00 out of 1.00 Flag question

Write a program that returns the sum of the last digit of the given number and the second last digit of the given number.

Note1 : If the given number is -107, the second last digit is 1.

Note2 : The second last digit should be returned as a positive number, i.e. if the given number is -717, the second last digit is 1.

Note3 : If the given number is 0, then also the second last digit is 0. In such case, the program should return 7, i.e. if the given number is 0, the second last digit should be returned as 7.

For example,

If the given number is 432, the sum is 5

If the given number is -107, the sum is 8

**For example:**

| Input | Result |
|---|---|
| 432 | 5 |
| -107 | 8 |
| 0 | 7 |
| -10 | 1 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
print(abs(n)%10+abs(n)//10%10)
```

Reset answer

Test | Input | Result
1 | 432 | 5
2 | -107 | 8
3 | 0 | 7
4 | -10 | 1

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 7** Correct - Mark 1.00 out of 1.00 Flag question

Write a program that returns the second last digit of the given number. Second last digit is being referred the digit in the ten's place in the given number.

For single digit numbers, the second last digit is 0.

Note1 : The second last digit should be returned as a positive number, i.e. if the given number is -717, the second last digit is 1.

Note2 : If the given number is 0, then also the second last digit is 0.

For example,

If the given number is 432, the second last digit is 3

If the given number is -107, the second last digit is 0

**For example:**

| Input | Result |
|---|---|
| 432 | 3 |
| -107 | 0 |
| 0 | 0 |
| -10 | 1 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
print(abs(n)//10%10)
```

Reset answer

Test | Input | Result
1 | 432 | 3
2 | -107 | 0
3 | 0 | 0
4 | -10 | 1

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 8** Correct - Mark 1.00 out of 1.00 Flag question

Write a program that returns the last digit of the given number.

Note1 : If the given number is -107, the second last digit is 1.

Note2 : The second last digit should be returned as a positive number, i.e. if the given number is -717, the second last digit is 1.

Note3 : If the given number is 0, then also the second last digit is 0. In such case, the program should return 7, i.e. if the given number is 0, the second last digit should be returned as 7.

For example,

If the given number is 432, the sum is 5

If the given number is -107, the sum is 8

**For example:**

| Input | Result |
|---|---|
| 432 | 2 |
| -107 | 7 |
| 0 | 7 |
| -10 | 0 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
print(abs(n)%10)
```

Reset answer

Test | Input | Result
1 | 432 | 2
2 | -107 | 7
3 | 0 | 7
4 | -10 | 0

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 9** Correct - Mark 1.00 out of 1.00 Flag question

Implement GPA calculator and GPA range and GPA of all Grand Average(GA), write a python code to print result list of boys, girls and Grand students in the specific format using multiple operator.

Input Format:

Input 1 Total Students : 1000, Boys : 450, Girls : 550, Result : Cut Total Students : 1000, Pass : 900, Girls : 900, Result : Cut

**For example:**

| Input | Result |
|---|---|
| 1000 1000 450 550 1000 900 900 | Total Students : 1000, Boys : 450, Girls : 550, Result : Cut<br>Total Students : 1000, Pass : 900, Girls : 900, Result : Cut |

**Answer:** (penalty regime: 0 %)

```
a=int(input())
b=int(input())
c=int(input())
print("Total Students :",a,"Boys :",b,"Girls :",c,"Result :","Cut")
```

Reset answer

Test | Input | Result
1 | 1000 1000 450 550 1000 900 900 | Total Students : 1000, Boys : 450, Girls : 550, Result : Cut<br>Total Students : 1000, Pass : 900, Girls : 900, Result : Cut

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 10** Correct - Mark 1.00 out of 1.00 Flag question

Write a program to print the alphabet like given input number is even.

If the given number is even, the program should return 2 rows in the pattern.

Note: The number passed to the program can either be negative, positive or zero, zero should be treated as even.

**For example:**

| Input | Result |
|---|---|
| 10 | abcdefghijk<br>abcdefghijk |
| -20 | abcdefghijklmnopqrst<br>abcdefghijklmnopqrst |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
if(n%2==0):
    for i in range(2):
        for j in range(97,97+n+1):
            print(chr(j),end="")
        print()
```

Reset answer

Test | Input | Result
1 | 10 | abcdefghijk<br>abcdefghijk
2 | -20 | abcdefghijklmnopqrst<br>abcdefghijklmnopqrst
3 | 0 | 
4 | 2 | abc<br>abc

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 11** Correct - Mark 1.00 out of 1.00 Flag question

Write a program that prints the given input number is even.

If the given number is even, the program should return 2 rows in the pattern.

Note: The number passed to the program can either be negative, positive or zero, zero should be treated as even.

**For example:**

| Input | Result |
|---|---|
| 10 | 10<br>10 |
| -20 | -20<br>-20 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
if(n%2==0):
    for i in range(2):
        print(n)
```

Reset answer

Test | Input | Result
1 | 10 | 10<br>10
2 | -20 | -20<br>-20
3 | 0 | 0<br>0
4 | 2 | 2<br>2

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 12** Correct - Mark 1.00 out of 1.00 Flag question

Write a program that prints the given input number is even.

If the given number is even, the program should return 2 rows in the pattern.

Note: The number passed to the program can either be negative, positive or zero, zero should be treated as even.

**For example:**

| Input | Result |
|---|---|
| 10 | 10<br>10 |
| -20 | -20<br>-20 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
if(n%2==0):
    for i in range(2):
        print(n)
```

Reset answer

Test | Input | Result
1 | 10 | 10<br>10
2 | -20 | -20<br>-20
3 | 0 | 0<br>0
4 | 2 | 2<br>2

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 13** Correct - Mark 1.00 out of 1.00 Flag question

In today's technology, a small object is added to certain consumers to encourage people to recycle them. In one particular jurisdiction, drink containers holding 100 ml or less, from 150 ml and 24 ml to containers holding more than one liter have a 10¢ deposit. What program that reads the number of containers returned by a customer and prints the refund owed. Assume that all containers are recyclable. Your program only successful receiving those containers. Finally the function is to print a message that reads right and says thanks have the refund owed.

Sample Input:
10
Sample Output:
Your total refund would be $1.00

**For example:**

| Input | Result |
|---|---|
| 10 | Your total refund will be $1.00<br>Thank you! |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
if(n<=10):
    print("Your total refund will be $1.00")
    print("Thank you!")
```

Reset answer

Test | Input | Result
1 | 10 | Your total refund will be $1.00<br>Thank you!
2 | 5 | Your total refund will be $1.00<br>Thank you!
3 | 0 | Your total refund will be $1.00<br>Thank you!
4 | 100 | Your total refund will be $1.00<br>Thank you!

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

## CD23321 Python Programming for Design-2025 Batch

Question 1 Correct Mark 10.00 out of 10.00 Flag question

Write a Python program that reads the word entered. The file is present in an image. The word is one of the following mentioned in parenthesis ( ) . If the first and last letter are the same in the word.

Print "First and last letter are same" else print "First and last letter are not same"

(Level, Python, Program, Code, Data, Input, Output, Variable, Function, Loop, Condition, Array, String, Integer, Float, Boolean, Print, Return, Error, Debug, Syntax, Indentation, Comment, Module, Import, Class, Object, Method, Attribute, Exception, File, Read, Write, Append, Dictionary, List, Tuple, Set, Key, Value, Index, Slice, Length, Type, Conversion, Operator, Expression, Statement, Assignment, Comparison, Logical, Arithmetic, Modulus, Power, Division, Multiplication, Addition, Subtraction, Parentheses, Order, Precedence, Associativity, Left, Right, Unary, Binary, Ternary, Short-circuit, Evaluation, Truthy, Falsy, None, Null, Undefined, Global, Local, Scope, Namespace, Recursion, Base case, Stack, Heap, Memory, Garbage collection, Reference, Mutable, Immutable, Hashable, Iterable, Iterator, Generator, Yield, Lambda, Map, Filter, Reduce, Zip, Enumerate, Sorted, Reversed, Any, All, Min, Max, Sum, Abs, Round, Pow, Divmod, Bin, Hex, Oct, Ord, Chr, Format, Repr, Str, Int, Float, Bool, List, Tuple, Set, Dict, Frozenset, Bytes, Bytearray, Memoryview, Complex, Range, Slice, Ellipsis, NotImplemented, Super, Property, Staticmethod, Classmethod, Abstract, Interface, Inheritance, Polymorphism, Encapsulation, Abstraction, Composition, Aggregation, Association, Dependency, Realization, Generalization, Specialization, Multiplicity, Navigability, Visibility, Public, Private, Protected, Package, Module, Import, Export, Namespace, Scope, Lifetime, Binding, Assignment, Declaration, Definition, Initialization, Constructor, Destructor, Garbage collection, Memory leak, Dangling pointer, Null pointer, Segmentation fault, Bus error, Floating point exception, Illegal instruction, Stack overflow, Heap overflow, Buffer overflow, Integer overflow, Underflow, Rounding error, Precision, Accuracy, Error handling, Exception handling, Try, Except, Finally, Raise, Assert, Warning, Logging, Debugging, Profiling, Testing, Unit testing, Integration testing, System testing, Acceptance testing, Regression testing, Performance testing, Load testing, Stress testing, Security testing, Usability testing, Compatibility testing, Portability testing, Maintainability testing, Reliability testing, Availability testing, Scalability testing, Flexibility testing, Extensibility testing, Reusability testing, Modularity testing, Cohesion, Coupling, Abstraction, Encapsulation, Inheritance, Polymorphism, Composition, Aggregation, Association, Dependency, Realization, Generalization, Specialization, Multiplicity, Navigability, Visibility, Public, Private, Protected, Package, Module, Import, Export, Namespace, Scope, Lifetime, Binding, Assignment, Declaration, Definition, Initialization, Constructor, Destructor, Garbage collection, Memory leak, Dangling pointer, Null pointer, Segmentation fault, Bus error, Floating point exception, Illegal instruction, Stack overflow, Heap overflow, Buffer overflow, Integer overflow, Underflow, Rounding error, Precision, Accuracy, Error handling, Exception handling, Try, Except, Finally, Raise, Assert, Warning, Logging, Debugging, Profiling, Testing, Unit testing, Integration testing, System testing, Acceptance testing, Regression testing, Performance testing, Load testing, Stress testing, Security testing, Usability testing, Compatibility testing, Portability testing, Maintainability testing, Reliability testing, Availability testing, Scalability testing, Flexibility testing, Extensibility testing, Reusability testing, Modularity testing, Cohesion, Coupling)

Input:
Level
Output:
First and last letter are same

Input:
Python
Output:
First and last letter are not same

Answer: (penalty regime: 0 %)

```
word = input()
if word[0] == word[-1]:
    print("First and last letter are same")
else:
    print("First and last letter are not same")
```

| Input | Expected | Got |
|---|---|---|
| Level | First and last letter are same | First and last letter are same |
| Python | First and last letter are not same | First and last letter are not same |

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

Question 2 Correct Mark 10.00 out of 10.00 Flag question

Write a program to display the following pattern using the user's name number of rows in pyramid.

Input Format

Input consists of a single integer n, the number of rows in the pyramid. Input is 5

Output Format

Output consists of n lines containing integers in Middle to Center to Side pattern.

Sample Input 1

5

Sample Output 1

Code Here

Answer: (penalty regime: 0 %)

```
n = int(input())
for i in range(1, n+1):
    print(" " * (n-i) + "* " * i)
for i in range(n-1, 0, -1):
    print(" " * (n-i) + "* " * i)
```

| Input | Expected | Got |
|---|---|---|
| 5 | * <br> * * <br> * * * <br> * * * * <br> * * * * * <br> * * * * <br> * * * <br> * * <br> * | * <br> * * <br> * * * <br> * * * * <br> * * * * * <br> * * * * <br> * * * <br> * * <br> * |

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

Question 3 Correct Mark 10.00 out of 10.00 Flag question

An Item

Write a Python program to create a class called Item with the following attributes:

- A name of the item and its price when the item is created.
- A method called total that takes a number of items as an argument. It should return the total cost of that number of items.
- A method called summary that takes no arguments. It should return a string that says "The name item costs $price each."

Input Format

Input consists of 2 integers

The first integer corresponds to the number of numbers given and the second integer corresponds to the number of numbers entered.

Output Format

Output consists of the string "Total = "and n

Sample Input and Output

Input:
5
3
Output:
Total = 15

Answer: (penalty regime: 0 %)

```
class Item:
    def __init__(self, name, price):
        self.name = name
        self.price = price
    def total(self, number):
        return number * self.price
    def summary(self):
        return f"The {self.name} item costs ${self.price}."
n = int(input())
m = int(input())
item = Item("Item", n)
print("Total =", item.total(m))
```

| Input | Expected | Got |
|---|---|---|
| 5<br>3 | Total = 15 | Total = 15 |
| 10<br>2 | Total = 20 | Total = 20 |
| 7<br>4 | Total = 28 | Total = 28 |
| 3<br>5 | Total = 15 | Total = 15 |
| 8<br>1 | Total = 8 | Total = 8 |

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

Question 4 Correct Mark 10.00 out of 10.00 Flag question

Write a program that accepts a sentence and calculate the number of letters and digits.

Input a string

Hello world! 123

Output

LETTERS 10

DIGITS 3

Sample Input and Output

Input:
Hello world! 123
Output:
LETTERS 10
DIGITS 3

Input:
Python 3.2
Output:
LETTERS 6
DIGITS 2

Answer: (penalty regime: 0 %)

```
s = input()
letters = 0
digits = 0
for c in s:
    if c.isalpha():
        letters += 1
    elif c.isdigit():
        digits += 1
print("LETTERS", letters)
print("DIGITS", digits)
```

| Input | Expected | Got |
|---|---|---|
| Hello world! 123 | LETTERS 10<br>DIGITS 3 | LETTERS 10<br>DIGITS 3 |
| Python 3.2 | LETTERS 6<br>DIGITS 2 | LETTERS 6<br>DIGITS 2 |

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

Question 5 Correct Mark 10.00 out of 10.00 Flag question

The Fibonacci series begins with numbers 0 and 1. Each term after the first two is the sum of the two previous terms. The series starts like this: 0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711 28657 46368 75025 121393 196418 317811 514229 832040 1346269 2178309 3524578 5702887 9227465 14930352 24157817 39088169 63245986 102334155 165580141 267914296 433494437 701408733 1134903170 1836311903 2971215073 4807526976 7778742049 12586269025 20365011074 32951280099 53316291173 86267571272 139583862445 225851433717 365435296162 591286729879 956722026041 1548008755920 2504730781961 4052739537881 6557470319842 10610209857723 17167680177565 27777890035288 44945570212853 72723460248141 117669030460994 190392490709135 308061521170129 498454011879264 806515533049393 1304969544928657 2111485077978050 3416454622906707 5527939700884757 8944394323791464 14472334024676221 23416728348467685 37889062373143906 61305790721611591 99194853094755497 160500643816367088 259695496911122585 420196140727489673 679891637638612258 1100087778366101931 1779979416004714189 2880067194370816120 4660046610375530309 7540113804746346429 12200160415121876738 19740274219868223167 31940434634990099905 51680708854858323072 83621143489848422977 135301852344706746049 218922995834555169026 354224848179261915075 573147844013817084101 927372692193078999176 1500520536206896083277 2427893228399975082453 3928413764606871165730 6356306993006846248183 10284720757613717413913 16641027750620563662096 26925748508234281076009 43566776258854844738105 70492524767089125814114 114059301025943970552219 184551825793033096366333 298611126818977066918552 483162952612010163284885 781774079430987230203437 1264937032042997393488322 2046711111473984623691759 3311648143516982017180081 5358359254990966640871840 8670007398507948658051921 14028366653498915298923761 22698374052006863956975682 36726740705505779255899443 59425114757512643212875125 96151855463018422468774568 155576970220531065681649693 251728825683549488150424261 407305795904080553832073954 659034621587630041982498215 1066340417491710595814572169 1725375039079340637797070384 2791715456571051233611642553 4517090495650391871408712937 7308805952221443105020355490 11825896447871834976429068427 19134702400093278081449423917 30960598847965113057878492344 50095301248058391139327916261 81055900096023504197206408605 131151201344081895336534324866 212207101440105399533740733471 343358302784187294870275058337 555565404224292694404015791808 898923707008479989274290850145 1454489111232772683678306641953 2353412818241252672952597492098 3807901929474025356630904134051 6161314747715278029583501626149 9969216677189303386214405760200 16130531424904581415797907386349 26099748102093884802012313146549 42230279526998466217810220532898 68330027629092351019822533679447 110560307156090817237632754212345 178890334785183168257455287891792 289450641941273985495088042104137 468340976726457153752543329995929 757791618667731139247631372100066 1226132595394188293000174702095995 1983924214061919432247806074196061 3210056809456107725247980776292056 5193981023518027157495786850488117 8404037832974134882743767626780173 13598018856492162040239554477268290 22002056689466296922983322104048463 35600075545958458963222876581316753 57602132235424755886206198685365216 93202207781383214849429075266681969 150804340016807970735635273952047185 244006547798191185585064349218729154 394810887814999156320699623170776339 638817435613190341905763972389505493 1033628323428189498226463595560281832 1672445759041379840132227567949787325 2706074082469569338358691163510069157 4378519841510949178490918731459856482 7084593923980518516849609894969925639 11463113765491467695340528626429782121 18547707689471986212190138521399707760 30010821454963453907530667147829489881 48558529144435440119720805669229197641 78569350599398894027251472817058687522 127127879743834334146972278486287885163 205697230343233228174223751303346572685 332825110087067562321196029789634457848 538522340430300790495419781092981030533 871347450517368352816615810882615488381 1409869790947669143312035591975596518914 2281217241465037496128651402858211507295 3691087032412706639440686994833808026209 5972304273877744135569338397692019533504 9663391306290450774010025392525827559713 15635695580168194909579363790217847093217 25299086886458645683589389182743674652930 40934782466626840593168753972961521746147 66233869353085486276758143155705196399077 107168651819712326869926897128666718145224 173402521172797813146685040284371914544301 280571172992510140016611937413038632689525 454023694165307953163296977697410547233826 734594867157818093179908915110449179923351 1188618561323126046343205892807859727157177 1923213428480944139523114807918308907080528 3111831989804070185866320700726168634237705 5035045418284014325389435508644477541318233 8146877408088084511255756209370646175555938 13181922826372098836645191718015123716874171 21328800234460183347900947927385769892430109 34510723060832282184546139645400893609304280 55839523295292465532447087572786663501734389 90350246356124747716993227218187557111038669 146189769651417213249440314790974220612773058 236540016007541960966433541009161777723811727 382729785658959174215873855800136000336584785 619269801666501135182307396809297778060396512 1002000587325460309398181252609433778396981297 1621270388991961444580488649418731556457377809 2623270976317421753978669902028165334854359106 4244541365309383198559158551446896891311736915 6867812341626804952537828453475062226166096021 11112353706936188151096987004921959117477832936 17980166048562993103634815458397021343643928957 29092519755499181254731802463318980461121761893 47072685804062174358366617921715001804765690850 76165205559561355613098420385033982265887452743 123237891363623529971465038306749984070653143593 199403096923184885584563458691783966336540596336 322641988286808415556028496998533950407193739929 522045085210993301140591955690317916743734336265 844687073497801716696620452688851867150928076194 1366732158708795017837212408379169783894662412459 2211419232206596734533832861067021651045590488653 3578151390915391752371045269446191434940252901112 5789570623121988486904878130513213085985843389765 9367722014037380239275923399959404520926096290877 15157292637159368726180801530472617606911939680642 24525014651196748965456724930431022127838036071519 39682307288356117691637526460903639734749975752161 64207321939552866657094251391334661862588011823680 103889629227908984348731777852238301597337987575841 168096951167461851005826029243572963459926999399521 272986580395370835354557807095811265057264986975362 441083531562832686360383836339384228517191986374883 714070111958203521714941643435195493574456973350245 1155153643521036208075325479774579722091648959725128 1869223755479239729790267123209775215666105933075373 3024377398900275937865592603984354937757754892800501 4893601154379515667655859727194130153423860825875874 7917978553279791605521452331178485091181615718676375 12811579707659307273177312058372615244605476544552249 20729558260938098878698764389551100335787092263228624 33541137968597406151876076447923715580392568807780873 54270696229535505030574840837474815916179661071009497 87811834198132911182450917285398531496572229878790370 142082530427668416212025758122873347412751890949799867 229894364625801327394476675408271878909324120828590237 372976895053469743606502433531145226322076011778390104 603871259679271070001978108939417105231400132606980341 977848154732740813608480542470562331553476144385370445 1581719414412011883610458651409979436784876276992350786 2559567569144752697218939193880541768338352421377721231 4141287083556764580829397845290521205123228698370072017 6700854652701517278048337039171062973461581119747793248 10842141736258281858877734884461584178584809818117865265 17542996388959799136926071923632647152046390937865658513 28385138125218080995803806808094231330631200755983523778 45928134514177880132729878731726878482677591693849182291 74313272639395961128533685539821109813308792449832706069 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# CD23321-Python Programming for Design-2025 Batch

| | |
|---|---|
| **Started on** | Tuesday, 4 August 2026, 1:10 PM |
| **State** | Finished |
| **Completed on** | Thursday, 6 August 2026, 4:46 PM |
| **Time taken** | 2 days 3 hours |
| **Marks** | 5.00/5.00 |
| **Grade** | **100.00** out of 100.00 |

**Quiz navigation**

1 2 3 4 5

Show one page at a time

Finish review

## Question 1

Correct Mark 1.00 out of 1.00 Flag question

Alice, an insurance agent, needs a program to calculate the insurance premium for her clients based on their age and health condition.

The premium amount is determined by the following rules:

1. If the client's age is between 18 and 30 years:
2. 'excellent' health condition: 500.0
3. 'good' health condition: 700.0
4. Any other health condition: 900.0
5. If the client's age is between 31 and 50 years:
6. 'excellent' health condition: 700.0
7. 'good' health condition: 900.0
8. Any other health condition: 1100.0

Write a program that takes the age and health condition of a client as input and outputs the corresponding insurance premium.

**Input Format :**
The first line of input contains an integer representing the age of the client.
The second line contains a string representing the health condition of the client.

**Output Format :**
The output prints a double value, representing the insurance premium.

**For example:**

| Input | Result |
|---|---|
| 25<br>excellent | 500.0 |
| 35<br>good | 900.0 |
| 45<br>poor | 1100.0 |

**Answer:** (penalty regime: 0 %)

```
age=int(input())
a=input()
if(18<=age<=30):
    if(a=="excellent"):
        print("500.0")
    elif(a=="good"):
        print("700.0")
    else:
        print("900.0")
elif
    if(a=="excellent"):
        print("700.0")
    elif(a=="good"):
        print("900.0")
    else:
        print("1100.0")
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 25<br>excellent | 500.0 | 500.0 | ✓ |
| ✓ | 35<br>good | 900.0 | 900.0 | ✓ |
| ✓ | 45<br>poor | 1100.0 | 1100.0 | ✓ |
| ✓ | 18<br>good | 900.0 | 900.0 | ✓ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

## Question 2

Correct Mark 1.00 out of 1.00 Flag question

Raor wants to estimate the total utility bill for a household based on the consumption of electricity, water, and gas.

Write a program to calculate the total bill using the following criteria:

1. The cost per unit for electricity is 0.12, for water is 0.05, and for gas is 0.08.
2. A discount is applied to the total cost based on the following conditions:
3. If the total cost is 100 or more, a 10% discount is applied.
4. If the total cost is between 50 and 100, a 5% discount is applied.
5. No discount is applied if the total cost is less than 50.

The program should output the total bill after applying the discount with two decimal places.

**Input Format :**
The input consists of three double values, representing the number of units consumed for electricity, water, and gas respectively.

**Output Format :**
The output prints a double value, representing the total bill after applying the discount, formatted to two decimal places.

**For example:**

| Input | Result |
|---|---|
| 1000.0 500.0 200.0 | 124.20 |
| 100.0 50.0 25.0 | 19.95 |
| 120.0 75.0 40.0 | 21.50 |

**Answer:** (penalty regime: 0 %)

```
a=int(input())
b=int(input())
c=int(input())
#
#
#
x=(a*0.12+b*0.05+c*0.08)
#
#
if(x>=100):
    a=(0.1*x)
    print("%.2f"%a)
elif(x>=50 and x<=100):
    a=(0.05*x)
    print("%.2f"%a)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 1000.0 500.0 200.0 | 124.20 | 124.20 | ✓ |
| ✓ | 100.0 50.0 25.0 | 19.95 | 19.95 | ✓ |
| ✓ | 120.0 75.0 40.0 | 21.50 | 21.50 | ✓ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

## Question 3

Correct Mark 1.00 out of 1.00 Flag question

Akash is tasked with developing a program that calculates and categorizes blood pressure based on the given systolic and diastolic readings.

The program should use the following classifications:

1. Low Blood Pressure: Systolic < 90 mm Hg or Diastolic < 60 mm Hg
2. Normal Blood Pressure: Systolic < 120 mm Hg and Diastolic ≤ 80 mm Hg
3. Prehypertension: Systolic ≤ 140 mm Hg and Diastolic ≤ 90 mm Hg
4. Stage 1 Hypertension: Systolic ≤ 160 mm Hg and Diastolic ≤ 100 mm Hg
5. Stage 2 Hypertension: Otherwise

Write a program to assist Akash in computing and classifying blood pressure levels based on input readings.

**Input Format :**
The input consists of two space-separated integers, representing the systolic blood pressure value **S** and diastolic blood pressure value **D**, respectively.

**Output Format :**
The output displays "Blood Pressure Category: " followed by the blood pressure category based on the provided input.

**Refer to the sample output for the exact text and format.**

**For example:**

| Input | Result |
|---|---|
| 90 85 | Blood Pressure Category: Low Blood Pressure |
| 110 70 | Blood Pressure Category: Normal Blood Pressure |

**Answer:** (penalty regime: 0 %)

```
s=int(input())
d=int(input())
#
#
if(s<90 or d<60):
    print("Blood Pressure Category: Low Blood Pressure")
elif(s<=120 and d<=80):
    print("Blood Pressure Category: Normal Blood Pressure")
elif(s<=140 and d<=90):
    print("Blood Pressure Category: Prehypertension")
elif(s<=160 and d<=100):
    print("Blood Pressure Category: Stage 1 Hypertension")
else:
    print("Blood Pressure Category: Stage 2 Hypertension")
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 90<br>85 | Blood Pressure Category: Low Blood Pressure | Blood Pressure Category: Low Blood Pressure | ✓ |
| ✓ | 110<br>70 | Blood Pressure Category: Normal Blood Pressure | Blood Pressure Category: Normal Blood Pressure | ✓ |
| ✓ | 120<br>80 | Blood Pressure Category: Prehypertension | Blood Pressure Category: Prehypertension | ✓ |
| ✓ | 145<br>98 | Blood Pressure Category: Stage 1 Hypertension | Blood Pressure Category: Stage 1 Hypertension | ✓ |
| ✓ | 170<br>110 | Blood Pressure Category: Stage 2 Hypertension | Blood Pressure Category: Stage 2 Hypertension | ✓ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

## Question 4

Correct Mark 1.00 out of 1.00 Flag question

Dinesh is developing a geometry calculator program for a Math class. He wants to write a program that allows users to calculate the area of different geometric shapes such as circles, rectangles, and triangles.

Users can select the type of shape they want to calculate the area for, and the program will perform the corresponding calculation using a 'switch-case' statement.

**Formulas Used:**

Area of Circle = 3.14 * radius * radius

Area of Rectangle = length * width

Area of Triangle = 0.5 * base * height

**Input Format :**
The first line of input consists of an integer, representing the geometric shape the user wants to calculate the area for (1 for a circle, 2 for a rectangle, and 3 for a triangle).

1. If the choice is 1, the second line of input consists of a double value **R**, representing the radius of the circle.
2. If the choice is 2, the second line and third lines of input consists of double values **L** and **B**, representing the length and width of the rectangle.
3. If the choice is 3, the second and third lines of input consists of double values **B** and **H**, representing the base and height of the triangle.

**Output Format :**
If the valid choice is entered, the output prints "**X sq.units**" where X represents the calculated area of the specific shape, rounded off to two decimal places.
Otherwise, the output prints "**Invalid choice**"

**For example:**

| Input | Result |
|---|---|
| 1<br>31.07 | 31.07 sq.units |
| 2<br>26.23<br>4.1 | 26.23 sq.units |
| 4 | Invalid choice |

**Answer:** (penalty regime: 0 %)

```
ch=int()
if(ch==1):
    r=float(input())
    a=3.14*r*r
    print("%.2f"%a,"sq.units")
elif(ch==2):
    l=float(input())
    b=float(input())
    a=l*b
    print("%.2f"%a,"sq.units")
elif(ch==3):
    b=float(input())
    h=float(input())
    a=0.5*b*h
    print("%.2f"%a,"sq.units")
else:
    print("Invalid choice")
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 1<br>31.07 | 31.07 sq.units | 31.07 sq.units | ✓ |
| ✓ | 2<br>26.23<br>4.1 | 26.23 sq.units | 26.23 sq.units | ✓ |
| ✓ | 3<br>24.56<br>5.2 | 24.56 sq.units | 24.56 sq.units | ✓ |
| ✓ | 4 | Invalid choice | Invalid choice | ✓ |
| ✓ | 1<br>265.58<br>9.5 | 265.58 sq.units | 265.58 sq.units | ✓ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

## Question 5

Correct Mark 1.00 out of 1.00 Flag question

Samantha is a diligent math student who is exploring the world of programming. She is learning Java and has recently studied conditional statements. One day, her teacher gives her an interesting problem to solve, that takes a number as input and checks whether it is a multiple of 5 or 7.

Help her to complete the task.

**Input Format :**
The input consists of a single integer **N**, representing the number to be checked.

**Output Format :**
If the number is a multiple of 5 but not 7, the output prints "N is a multiple of 5".
If the number is a multiple of 7, the output prints "N is a multiple of 7".
Otherwise, the output prints "N is neither multiple of 5 nor 7" where N is an entered integer.

**For example:**

| Input | Result |
|---|---|
| 10 | 10 is a multiple of 5 |
| 21 | 21 is a multiple of 7 |
| 11 | 11 is neither multiple of 5 nor 7 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
#
if(n%5==0):
    print("%d is a multiple of 5"%n)
elif(n%7==0):
    print("%d is a multiple of 7"%n)
else:
    print("%d is neither multiple of 5 nor 7"%n)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 10 | 10 is a multiple of 5 | 10 is a multiple of 5 | ✓ |
| ✓ | 21 | 21 is a multiple of 7 | 21 is a multiple of 7 | ✓ |
| ✓ | 11 | 11 is neither multiple of 5 nor 7 | 11 is neither multiple of 5 nor 7 | ✓ |
| ✓ | 9 | 9 is neither multiple of 5 nor 7 | 9 is neither multiple of 5 nor 7 | ✓ |
| ✓ | 70 | 70 is a multiple of 7 | 70 is a multiple of 7 | ✓ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

# CD23321-Python Programming for Design-2025 Batch

| | |
|---|---|
| Started on | Saturday, 9 August 2025, 7:30 AM |
| State | Finished |
| Completed on | Saturday, 9 August 2025, 10:19 PM |
| Time taken | 14 hours 49 mins |
| Grade | **100.00** out of 100.00 |

**Question 1** Correct Mark 100.00 out of 100.00 Flag question

Find how many subsets in an array such that the sum of all the elements in each subset is divisible by k.

Example 1:

Input: nums = [4,5,0,-2,-3,1], k = 5

Output: 7

Explanation: There are 7 subsets with a sum divisible by k = 5: [5], [5, 0], [5, 0, -2, -3], [0], [0, -2, -3], [-2, -3]

Example 2:

Input: nums = [5], k = 9

Output: 0

Explanation: No subset has a sum divisible by k = 9.

Constraints:

1 <= nums.length <= 30000

-10000 <= nums[i] <= 10000

2 <= k <= 10000

**For example:**

| Input | Result |
|---|---|
| 6<br>4 5 0 -2 -3 1<br>5 | 7 |
| 1<br>5<br>9 | 0 |

**Answer:** (penalty regime: 0 %)

```
Reset answer
```

**Question 2** Correct Mark 100.00 out of 100.00 Flag question

Write a program to find the number of non-repeated digits in given number N. The number will be passed to the program as an input of type int.

Input format:

Input consists of a single integer number which is greater than or equal to 1 and <=10000.

Output format:

If the given number is 0, the program should return 1 because there is only 1 non-repeated digit in this number.

If the given number is 10, the program should return 10 because there are 2 non-repeated digits in this number, '1' and '0'.

If the given number is 101, the program should return 2 because there are 2 non-repeated digits in this number, '1' and '0'.

If the given number is 121, the program should return 2 because there are 2 non-repeated digits in this number, '1' and '2'.

**For example:**

| Input | Result |
|---|---|
| 0 | 1 |
| 10 | 10 |
| 101 | 2 |
| 121 | 2 |

**Answer:** (penalty regime: 0 %)

Sorry, only JAVA, Python and C++ are supported.

```
Reset answer
```

**Question 3** Correct Mark 100.00 out of 100.00 Flag question

Write a function to find the nth number that contains only the digits 4 and 7. Numbers must be sorted in ascending order.

Input Format

Integer input from stdin

Output Format

Print a single integer denoting the nth lucky number.

Example Input

11

Example Output

74

**For example:**

| Input | Result |
|---|---|
| 11 | 74 |

**Answer:** (penalty regime: 0 %)

Sorry, only JAVA, Python and C++ are supported.

```
Reset answer
```

**Question 4** Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern based on input size.

Input

5

Output

1

2 3

4 5 6

7 8 9 10

**For example:**

| Input | Result |
|---|---|
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

**Answer:** (penalty regime: 0 %)

Sorry, only JAVA, Python and C++ are supported.

```
Reset answer
```

**Question 5** Correct Mark 100.00 out of 100.00 Flag question

Write a program to find the number of prime numbers in a specified range.

The starting and ending number of the range will be provided as input to the program.

The program should return the count of prime numbers in the range as output of the program.

Example 1: If the starting and ending numbers of the range given as input are 10 & 20, the program must return 4, because there are 4 prime numbers in the specified range. Prime numbers in the range 10 to 20 are 11, 13, 17, 19.

Example 2: If the starting and ending numbers of the range given as input are 100 and 200, the program must return 21, because there are 21 prime numbers in the specified range. Prime numbers in the range 100 to 200 are 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199

**For example:**

| Input | Result |
|---|---|
| 10<br>20 | 4 |
| 100<br>200 | 21 |

**Answer:** (penalty regime: 0 %)

Sorry, only JAVA, Python and C++ are supported.

```
Reset answer
```

**Question 6** Correct Mark 100.00 out of 100.00 Flag question

Write a program to find the number of prime numbers in a specified range.

The starting and ending number of the range will be provided as input to the program.

The program should return the count of prime numbers in the range as output of the program.

Example 1: If the starting and ending numbers of the range given as input are 10 & 20, the program must return 4, because there are 4 prime numbers in the specified range. Prime numbers in the range 10 to 20 are 11, 13, 17, 19.

Example 2: If the starting and ending numbers of the range given as input are 100 and 200, the program must return 21, because there are 21 prime numbers in the specified range. Prime numbers in the range 100 to 200 are 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199

**For example:**

| Input | Result |
|---|---|
| 10<br>20 | 4 |
| 100<br>200 | 21 |

**Answer:** (penalty regime: 0 %)

Sorry, only JAVA, Python and C++ are supported.

```
Reset answer
```

**Question 7** Correct Mark 100.00 out of 100.00 Flag question

Write a program to find the number of prime numbers in a specified range.

The starting and ending number of the range will be provided as input to the program.

The program should return the count of prime numbers in the range as output of the program.

Example 1: If the starting and ending numbers of the range given as input are 10 & 20, the program must return 4, because there are 4 prime numbers in the specified range. Prime numbers in the range 10 to 20 are 11, 13, 17, 19.

Example 2: If the starting and ending numbers of the range given as input are 100 and 200, the program must return 21, because there are 21 prime numbers in the specified range. Prime numbers in the range 100 to 200 are 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199

**For example:**

| Input | Result |
|---|---|
| 10<br>20 | 4 |
| 100<br>200 | 21 |

**Answer:** (penalty regime: 0 %)

Sorry, only JAVA, Python and C++ are supported.

```
Reset answer
```

**Question 8** Correct Mark 100.00 out of 100.00 Flag question

Given a positive integer N, find the smallest positive integer M which is Multiple of both N and M.

A Simple Is to answer is the answer is not M in 1D but ignore Integer, then return M.

**Example 1**

Input

10

Output

10

**Example 2**

Input

15

Output

15

**For example:**

| Input | Result |
|---|---|
| 10 | 10 |
| 15 | 15 |

**Answer:** (penalty regime: 0 %)

Sorry, only JAVA, Python and C++ are supported.

```
Reset answer
```

**Question 9** Correct Mark 100.00 out of 100.00 Flag question

An automorphic number is a number whose square ends with the number itself.

For example, 5 is an automorphic number because 25 ends with 5. The first eight (8) numbers are:

0 1 5 6 25 76 376 625

Write a program to print the first N(N<=10) Automorphic numbers. Where N is the input.

Input Format

Takes a integer input from keyboard

Output Format

Print the Automorphic integer number in Automorphic number. otherwise Not Automorphic

Example Input

5

Output

Automorphic

Example Input

7

Output

Automorphic

Example Input

abc

Output

Invalid

**For example:**

| Input | Result |
|---|---|
| 5 | Automorphic |
| 7 | Automorphic |
| abc | Invalid |

**Answer:** (penalty regime: 0 %)

Sorry, only JAVA, Python and C++ are supported.

```
Reset answer
```

**Question 10** Correct Mark 100.00 out of 100.00 Flag question

Write a program that given an integer 'n', prints the number of integers that are less than or equal to 'n' and are a prime to 'n'.

Two integers a and b are said to be relatively prime or co-prime if the only positive integer that evenly divides both of them is 1. That is, the greatest common divisor of a and b is 1. For example, 14 and 15 are relatively prime, but 14 and 21 are not.

Input Format

Write the containing the value of 'n', where 1<=n<=10000

Output Format

Write the containing the number of integers that are co-prime to n and less than or equal to 'n'

Sample Test Cases

Test Case 1

Input

10

Output

4

Test Case 2

Input

97

Output

72

Test Case 3

Input

1

Output

1

**For example:**

| Input | Result |
|---|---|
| 10 | 4 |
| 97 | 72 |
| 1 | 1 |

**Answer:** (penalty regime: 0 %)

Sorry, only JAVA, Python and C++ are supported.

```
Reset answer
```

**Question 11** Correct Mark 100.00 out of 100.00 Flag question

You are choosing what size of shoe will serve a friend. For one pair, you decided that (assigned as) a number that looks to you like a shoe size.

The first integer represents the number of pairs of shoes you have to choose from.

The following integers are the shoe sizes of each pair of shoes, in order.

The last integer is your friend's assigned size for which you want to find the number of pairs of shoes that fit your friend's size.

**Input Format**

1st number of integers n

n number of integers

n number of integers

1 integer of integers

**Output Format**

Both integers are on the same position within 0 pairs, 1D, otherwise NO.

**For example:**

| Input | Result |
|---|---|
| 6<br>2<br>3<br>4<br>5<br>6<br>7<br>4 | 2 |

**Answer:** (penalty regime: 0 %)

```
Reset answer
```

**Question 12** Correct Mark 100.00 out of 100.00 Flag question

Write a program that given an integer 'n', prints the number of integers that are less than or equal to 'n' and are a prime to 'n'.

Two integers a and b are said to be relatively prime or co-prime if the only positive integer that evenly divides both of them is 1. That is, the greatest common divisor of a and b is 1. For example, 14 and 15 are relatively prime, but 14 and 21 are not.

Input Format

Write the containing the value of 'n', where 1<=n<=10000

Output Format

Write the containing the number of integers that are co-prime to n and less than or equal to 'n'

Sample Test Cases

Test Case 1

Input

10

Output

4

Test Case 2

Input

97

Output

72

Test Case 3

Input

1

Output

1

**For example:**

| Input | Result |
|---|---|
| 10 | 4 |
| 97 | 72 |
| 1 | 1 |

**Answer:** (penalty regime: 0 %)

```
Reset answer
```

**Question 13** Correct Mark 100.00 out of 100.00 Flag question

You are choosing what size of shoe will serve a friend. For one pair, you decided that (assigned as) a number that looks to you like a shoe size.

The first integer represents the number of pairs of shoes you have to choose from.

The following integers are the shoe sizes of each pair of shoes, in order.

The last integer is your friend's assigned size for which you want to find the number of pairs of shoes that fit your friend's size.

**Input Format**

1st number of integers n

n number of integers

n number of integers

1 integer of integers

**Output Format**

Both integers are on the same position within 0 pairs, 1D, otherwise NO.

**For example:**

| Input | Result |
|---|---|
| 6<br>2<br>3<br>4<br>5<br>6<br>7<br>4 | 2 |

**Answer:** (penalty regime: 0 %)

```
Reset answer
```

# CD23321-Python Programming for Design-2025 Batch

| | |
|---|---|
| Started on | Monday, 10 August 2026, 10:23 PM |
| State | Finished |
| Completed on | Monday, 10 August 2026, 11:34 PM |
| Time taken | 1 hour 10 mins |
| Marks | 10.00/10.00 |
| Grade | 100.00 out of 100.00 |



## Question 1
Correct Mark 1.00 out of 1.00 Flag question

Given an integer `num`, repeatedly add all its digits until the result has only one digit, and return it.

**Example 1:**

```
Input: num = 38
Output: 2
Explanation: The process is
38 --> 3 + 8 --> 11
11 --> 1 + 1 --> 2
Since 2 has only one digit, return it.
```

**Example 2:**

```
Input: num = 0
Output: 0
```

**For example:**

| Input | Result |
|---|---|
| 38 | 2 |
| 0 | 0 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())

while(n>=10):
    total=0
    while(n>0):
        total+=n%10
        n//=10
    n=total
print(n)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 38 | 2 | 2 | ✔ |
| ✔ | 0 | 0 | 0 | ✔ |
| ✔ | 11 | 2 | 2 | ✔ |
| ✔ | 50 | 5 | 5 | ✔ |
| ✔ | 91 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct
Marks for this submission: 1.00/1.00.

## Question 2
Correct Mark 1.00 out of 1.00 Flag question

Write a program in Python to display a pyramid with "*" as follows,

**For example:**

| Input | Result |
|---|---|
| 4 | *<br>***<br>*****<br>******* |

**Answer:** (penalty regime: 0 %)

```
n=int(input())

for i in range(1,n+1):
    sp=" "*(n-i)
    st="*"*(2*i-1)
    print(sp+st)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4 | *<br>***<br>*****<br>******* | *<br>***<br>*****<br>******* | ✔ |
| ✔ | 2 | *<br>*** | *<br>*** | ✔ |
| ✔ | 5 | *<br>***<br>*****<br>*******<br>********* | *<br>***<br>*****<br>*******<br>********* | ✔ |

Passed all tests! ✔

Correct
Marks for this submission: 1.00/1.00.

## Question 3
Correct Mark 1.00 out of 1.00 Flag question

Given a positive integer `n`, write a function that returns the number of `set bits` in its binary representation (also known as the Hamming weight).

**Example 1:**

**Input:** n = 11

**Output:** 3

**Explanation:**

The input binary string **1011** has a total of three set bits.

**Example 2:**

**Input:** n = 128

**Output:** 1

**Explanation:**

The input binary string **10000000** has a total of one set bit.

**Example 3:**

**Input:** n = 2147483645

**Output:** 30

**Explanation:**

The input binary string **1111111111111111111111111111101** has a total of thirty set bits.

**For example:**

| Input | Result |
|---|---|
| 11 | 3 |
| 128 | 1 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())

c=0
while n>0:
    c+=n%2
    n//=2
print(c)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11 | 3 | 3 | ✔ |
| ✔ | 128 | 1 | 1 | ✔ |
| ✔ | 32 | 1 | 1 | ✔ |
| ✔ | 2147483645 | 30 | 30 | ✔ |

Passed all tests! ✔

Correct
Marks for this submission: 1.00/1.00.

## Question 4
Correct Mark 1.00 out of 1.00 Flag question

You are climbing a staircase. It takes `n` steps to reach the top.

Each time you can either climb `1` or `2` steps. In how many distinct ways can you climb to the top?

**Example 1:**

```
Input: n = 2
Output: 2
Explanation: There are two ways to climb to the top.
1. 1 step + 1 step
2. 2 steps
```

**Example 2:**

```
Input: n = 3
Output: 3
Explanation: There are three ways to climb to the top.
1. 1 step + 1 step + 1 step
2. 1 step + 2 steps
3. 2 steps + 1 step
```

**Constraints:**

- `1 <= n <= 45`

**For example:**

| Input | Result |
|---|---|
| 2 | 2 |
| 3 | 3 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
a=1
b=2

for i in range(3,n+1):
    a,b=b,a+b

if(n>=2):
    print(b)
else:
    print(a)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 2 | 2 | 2 | ✔ |
| ✔ | 3 | 3 | 3 | ✔ |
| ✔ | 4 | 5 | 5 | ✔ |
| ✔ | 5 | 8 | 8 | ✔ |

Passed all tests! ✔

Correct
Marks for this submission: 1.00/1.00.

## Question 5
Correct Mark 1.00 out of 1.00 Flag question

Determine the factors of a number (i.e., all positive integer values that evenly divide into a number) and then return the $p^{th}$ element of the list, sorted ascending. If there is no $p^{th}$ element, return 0.

**Example**

n = 20

p = 3

The factors of 20 in ascending order are {1, 2, 4, 5, 10, 20}. Using 1-based indexing, if p = 3, then 4 is returned. If p > 6, 0 would be returned.

**Constraints**

$1 \le n \le 10^{15}$

$1 \le p \le 10^{9}$

The first line contains an integer n, the number to factor.

The second line contains an integer p, the 1-based index of the factor to return.

**Sample Case 0**

**Sample Input 0**

10

3

**Sample Output 0**

5

**Explanation 0**

Factoring n = 10 results in {1, 2, 5, 10}. Return the p = $3^{rd}$ factor, 5, as the answer.

**Sample Case 1**

**Sample Input 1**

10

5

**Sample Output 1**

0

**Explanation 1**

Factoring n = 10 results in {1, 2, 5, 10}. There are only 4 factors and p = 5, therefore 0 is returned as the answer.

**Sample Case 2**

**Sample Input 2**

1

1

**Sample Output 2**

1

**Explanation 2**

Factoring n = 1 results in {1}. The p = 1st factor of 1 is returned as the answer.

**For example:**

| Input | Result |
|---|---|
| 10<br>3 | 5 |
| 10<br>5 | 0 |
| 1<br>1 | 1 |

**Answer:** (penalty regime: 0 %)

```
n = int(input())
pos = int(input())

flag =0
cnt = 0

for i in range(1, n+1):
    if n%i==0:
        cnt+=1
        if cnt == pos:
            print(i)
            flag = 1
            break

if not (flag):
    print(0)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 10<br>3 | 5 | 5 | ✔ |
| ✔ | 10<br>5 | 0 | 0 | ✔ |
| ✔ | 1<br>1 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct
Marks for this submission: 1.00/1.00.

## Question 6
Correct Mark 1.00 out of 1.00 Flag question

A **happy number** is a number defined by the following process:

- Starting with any positive integer, replace the number by the sum of the squares of its digits.
- Repeat the process until the number equals 1 (where it will stay), or it **loops endlessly in a cycle** which does not include 1.
- Those numbers for which this process **ends in 1** are happy.

Print `true` *if* `n` *is a happy number, and* `false` *if not.*

**For example:**

| Input | Result |
|---|---|
| 19 | True |
| 2 | False |

**Answer:** (penalty regime: 0 %)

```
n=int(input())

while(n!=1 and n!=4):
    total=0

    while(n>0):
        d=n%10
        total+=d*d
        n//=10
    n=total
print(n==1)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 19 | True | True | ✔ |
| ✔ | 2 | False | False | ✔ |
| ✔ | 82 | True | True | ✔ |
| ✔ | 16 | False | False | ✔ |

Passed all tests! ✔

Correct
Marks for this submission: 1.00/1.00.

## Question 7
Correct Mark 1.00 out of 1.00 Flag question

A **perfect number** is a **positive integer** that is equal to the sum of its **positive divisors**, excluding the number itself. A **divisor** of an integer `x` is an integer that can divide `x` evenly.

Given an integer `n`, return `True` *if* `n` *is a perfect number, otherwise return* `False`.

**Example 1:**

```
Input: num = 28
Output: True
Explanation: 28 = 1 + 2 + 4 + 7 + 14
1, 2, 4, 7, and 14 are all divisors of 28.
```

**Example 2:**

```
Input: num = 7
Output: False
```

**Constraints:**

- `1 <= num <= 10`$^{8}$

**Answer:** (penalty regime: 0 %)

```
n=int(input())

total=0

for i in range(1, n):
    if n%i==0:
        total+=i
print(total==n)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 28 | True | True | ✔ |
| ✔ | 7 | False | False | ✔ |
| ✔ | 8128 | True | True | ✔ |
| ✔ | 496 | True | True | ✔ |
| ✔ | 500 | False | False | ✔ |

Passed all tests! ✔

Correct
Marks for this submission: 1.00/1.00.

## Question 8
Correct Mark 1.00 out of 1.00 Flag question

An **ugly number** is a *positive* integer which does not have a prime factor other than 2, 3, and 5.

Given an integer `n`, Print `True` *if* `n` *is an **ugly number,** Otherwise Print False.*

**For example:**

| Input | Result |
|---|---|
| 6 | True |
| 14 | False |

**Answer:** (penalty regime: 0 %)

```
n=int(input())

if(n<=0):
    print(False)
else:
    while(n%2==0):
        n//=2
    while(n%3==0):
        n//=3
    while(n%5==0):
        n//=5

print(n==1)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 6 | True | True | ✔ |
| ✔ | 14 | False | False | ✔ |
| ✔ | 125 | True | True | ✔ |
| ✔ | 21 | False | False | ✔ |

Passed all tests! ✔

Correct
Marks for this submission: 1.00/1.00.

## Question 9
Correct Mark 1.00 out of 1.00 Flag question

Given an integer `num`, return *the number of digits in* `num` *that divide* `num`.

An integer `val` divides `nums` if `nums % val == 0`.

**Example 1:**

```
Input: num = 7
Output: 1
Explanation: 7 divides itself, hence the answer is 1.
```

**Example 2:**

```
Input: num = 121
Output: 2
Explanation: 121 is divisible by 1, but not 2. Since 1 occurs twice as a digit, we return 2.
```

**Example 3:**

```
Input: num = 1248
Output: 4
Explanation: 1248 is divisible by all of its digits, hence the answer is 4.
```

**For example:**

| Input | Result |
|---|---|
| 7 | 1 |
| 121 | 2 |
| 1248 | 4 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
temp=n
c=0

while(temp>0):
    d=temp%10

    if(d!=0 and n%d==0):
        c+=1

    temp//=10
print(c)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 7 | 1 | 1 | ✔ |
| ✔ | 121 | 2 | 2 | ✔ |
| ✔ | 1248 | 4 | 4 | ✔ |
| ✔ | 12 | 2 | 2 | ✔ |
| ✔ | 45 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct
Marks for this submission: 1.00/1.00.

## Question 10
Correct Mark 1.00 out of 1.00 Flag question

Given an integer `n`, return *the number of trailing zeroes in* `n!`.

Note that `n! = n * (n - 1) * (n - 2) * ... * 3 * 2 * 1`.

**Example 1:**

```
Input: n = 3
Output: 0
Explanation: 3! = 6, no trailing zero.
```

**Example 2:**

```
Input: n = 5
Output: 1
Explanation: 5! = 120, one trailing zero.
```

**Example 3:**

```
Input: n = 0
Output: 0
```

**Constraints:**

- `0 <= n <= 10`$^{4}$

**For example:**

| Input | Result |
|---|---|
| 3 | 0 |
| 5 | 1 |

**Answer:** (penalty regime: 0 %)

```
n=int(input())
c=0

while(n>=5):
    n//=5
    c+=n
print(c)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3 | 0 | 0 | ✔ |
| ✔ | 5 | 1 | 1 | ✔ |
| ✔ | 0 | 0 | 0 | ✔ |
| ✔ | 10 | 2 | 2 | ✔ |
| ✔ | 25 | 6 | 6 | ✔ |

Passed all tests! ✔

Correct
Marks for this submission: 1.00/1.00.

# CD23321-Python Programming for Design-2025 Batch

| | |
|---|---|
| Started on | Tuesday, 18 August 2026, 12:03 PM |
| State | Finished |
| Completed on | Thursday, 20 August 2026, 4:23 PM |
| Time taken | 2 days 4 hours |
| Marks | 5.00/5.00 |
| Grade | 100.00 out of 100.00 |

Quiz navigation

1 2 3 4 5

Show one page at a time

Finish review

## Question 1

Correct Mark 1.00 out of 1.00 Flag question

A number is considered to be ugly if its only prime factors are 2, 3 or 5.

[1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, ...] is the sequence of ugly numbers.

Task:

complete the function which takes a number n as input and checks if it's an ugly number.

return ugly if it is ugly, else return not ugly

Hint:

An ugly number U can be expressed as: U = 2^a * 3^b * 5^c, where a, b and c are nonnegative integers.

**For example:**

| Test | Result |
|---|---|
| print(checkUgly(6)) | ugly |
| print(checkUgly(21)) | not ugly |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def checkUgly(n):
    while(n%2==0):
        n=n/2
    while(n%3==0):
        n=n/3
    while(n%5==0):
        n=n/5
    if(n==1):
        return "ugly"
    else:
        return "not ugly"
```

| Test | Expected | Got | |
|---|---|---|---|
| print(checkUgly(6)) | ugly | ugly | ✓ |
| print(checkUgly(21)) | not ugly | not ugly | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 2

Correct Mark 1.00 out of 1.00 Flag question

An e-commerce company plans to give their customers a special discount for Christmas.

They are planning to offer a flat discount. The discount value is calculated as the sum of all

the prime digits in the total bill amount.

Write an algorithm to find the discount value for the given total bill amount.

Constraints:

1 <= orderValue< 10e100000

Input

The input consists of an integer orderValue, representing the total bill amount.

Output

Print an integer representing the discount value for the given total bill amount.

Example Input

578

Output

12

**For example:**

| Test | Result |
|---|---|
| print(ChristmasDiscount(578)) | 12 |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def ChristmasDiscount(n):
    sum=0
    pos
    while(n>0):
        a=n%10
        if(a==2 or a==3 or a==5 or a==7):
            sum=sum+a
        n=n//10
    return sum
```

| Test | Expected | Got | |
|---|---|---|---|
| print(ChristmasDiscount(578)) | 12 | 12 | ✓ |
| print(ChristmasDiscount(57)) | 12 | 12 | ✓ |
| print(ChristmasDiscount(222)) | 6 | 6 | ✓ |
| print(ChristmasDiscount(7777771)) | 35 | 35 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 3

Correct Mark 1.00 out of 1.00 Flag question

complete function to implement coin change making problem i.e. finding the minimum

number of coins of certain denominations that add up to given amount of money.

The only available coins are of values 1, 2, 3, 4

Input Format:

Integer input from stdin.

Output Format:

return the minimum number of coins required to meet the given target.

Example Input:

16

Output:

4

Explanation:

We need only 4 coins of value 4 each

Example Input:

25

Output:

7

Explanation:

We need 6 coins of 4 value, and 1 coin of 1 value

**Answer:** (penalty regime: 0 %)

Reset answer

```
def coinChange(n):
    sum=0
    count=0
    while(sum<n):
        if(sum<n-4):
            sum=sum+4
            count+=1
        elif(sum<n-3):
            sum=sum+3
            count+=1
        elif(sum<n-2):
            sum=sum+2
            count+=1
        elif(sum<n-1):
            sum=sum+1
            count+=1
    return(count)
```

| Test | Expected | Got | |
|---|---|---|---|
| print(coinChange(16)) | 4 | 4 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 4

Correct Mark 1.00 out of 1.00 Flag question

A strobogrammatic number is a number that looks the same when rotated 180 degrees (looked at upside down).

Write a program to determine if a number is strobogrammatic. The number is represented as a string.

**Example 1:**

**Input:**

69

**Output:**

true

**Example 2:**

**Input:**

88

**Output:**

true

**Example 3:**

**Input:**

962

**Output:**

false

**Example 4:**

**Input:**

1

**Output:**

true

**For example:**

| Test | Result |
|---|---|
| print(StroboGrammatic(69)) | True |
| print(StroboGrammatic(962)) | False |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def StroboGrammatic(n):
    n=str(n)
    d={'0':'0','1':'1', '6':'9', '8':'8', '9':'6'}
    for i in range (len(n)):
        if n[i] not in d:
            return False
        if d[n[i]]!=n[len(n)-i-1]:
            return False
    return True

n=int(input())
print(StroboGrammatic(n))
```

| Test | Expected | Got | |
|---|---|---|---|
| print(StroboGrammatic(69)) | True | True | ✓ |
| print(StroboGrammatic(88)) | True | True | ✓ |
| print(StroboGrammatic(962)) | False | False | ✓ |
| print(StroboGrammatic(1)) | False | False | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 5

Correct Mark 1.00 out of 1.00 Flag question

Write a function that returns the value of a+aa+aaa+aaaa with a given digit as the value of a.

Suppose the following input is supplied to the program:

9

Then, the output should be:

9+99+999+9999=11106

Sample Input Format:

9

Sample Output Format:

11106

**For example:**

| Test | Result |
|---|---|
| print(Summation(9)) | 9872 |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def Summation(n):

    n= str(n)

    sum= int(n*1)+int(n*2)+int(n*3)+int(n*4)
    return sum
```

| Test | Expected | Got | |
|---|---|---|---|
| print(Summation(9)) | 9872 | 9872 | ✓ |
| print(Summation(10)) | 10280960 | 10280960 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Finish review

# CD23321-Python Programming for Design-2025 Batch

| | |
|---|---|
| **Started on** | Thursday, 20 August 2026, 4:26 PM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 4:07 PM |
| **Time taken** | 6 days 23 hours |
| **Marks** | 5.00/5.00 |
| **Grade** | **100.00** out of 100.00 |

**Quiz navigation**

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Finish review

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

Given an integer number and you have to count the digits using recursion using Python program. In this program, you will be reading an integer number and counting the total digits, using a function countDigits() which will take a number as an argument and return the count after recursion process.
Input Format: The first and only line of the input contains a single integer n
Output Format: Output a single line denoting the number of digits in n.

**For example:**

| Test | Result |
|---|---|
| `print(countDigits(3001))` | 3 |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def countDigits(n):
    count=0
    while(n>0):
        n=n//10
        count+=1
    return count
```

| | Test | Expected | Got |
|---|---|---|---|
| ✓ | `print(countDigits(1234567))` | 5 | 5 ✓ |
| ✓ | `print(countDigits(300))` | 3 | 3 ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

**Question 2** Correct Mark 1.00 out of 1.00 Flag question

Euclid was a Greek mathematician who lived approximately 2,300 years ago. His algorithm for computing the greatest common divisor of two positive integers, a and b, is both efficient and recursive. It is outlined below:

If b is 0 then
  return a
Else
  Set c equal to the remainder when a is divided by b
  Return the greatest common divisor of b and c

Write a Recursive function that implements Euclid's algorithm and uses it to determine the greatest common divisor of two integers entered by the user. Test your program with some very large integers. The result will be computed quickly, even for huge numbers consisting of hundreds of digits, because Euclid's algorithm is extremely efficient.

**For example:**

| Test | Result |
|---|---|
| `print(gcd(8, 12))` | 4 |
| `print(gcd(720, 1000))` | 40 |

**Answer:** (penalty regime: 0 %)

Reset answer

```
import math
def gcd(a,b):
    result=math.gcd(a,b)
    return result
```

| | Test | Expected | Got |
|---|---|---|---|
| ✓ | `print(gcd(8, 12))` | 4 | 4 ✓ |
| ✓ | `print(gcd(720, 1000))` | 40 | 40 ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

**Question 3** Correct Mark 1.00 out of 1.00 Flag question

The notion of a palindrome was introduced previously. In this exercise you will write a recursive function that determines whether or not a string is a palindrome. The empty string is a palindrome, as is any string containing only one character. Any longer string is a palindrome if its first and last characters match, and if the string formed by removing the first and last characters is also a palindrome.

Write a program that reads a string from the user and uses your recursive function to determine whether or not it is a palindrome. Then your program should display an appropriate message for the user.

Sample Input
malayalam
Sample Output
That was a palindrome!
Sample Input
madam
Sample Output
That is not a palindrome.

**Answer:** (penalty regime: 0 %)

Reset answer

```
def isPalindrome(s):
    if len(s) <= 1:
        return 1
    else:
        s=s[1:-1]

    if(s==s):
        return 1
    else:
        return 0
line=input()
if isPalindrome(line):
    print("That was a palindrome!")
else:
    print("That is not a palindrome.")
```

| | Input | Expected | Got |
|---|---|---|---|
| ✓ | malayalam | That was a palindrome! | That was a palindrome! ✓ |
| ✓ | madam | That is not a palindrome. | That is not a palindrome. ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

**Question 4** Correct Mark 1.00 out of 1.00 Flag question

Complete a Recursive Function to find if a given number N can be expressed as a sum of two prime numbers.
Note: YOU MUST OPTIMIZE the logic to find whether a number is prime or not, as very large prime numbers are provided as input. If the logic is not optimized your program will NOT get executed within the given time limit.

**Input Format:**

First line contains number N.

**Output Format:**

Return either yes or no.

**Boundary Conditions / Constraints:**

3 <= N <= 10^9

**Example Input/Output 1:**

**Input:**

20

**Output:**

yes

**Input:**

23

**Ouput:**

no

**Explanation:**

20 can be expressed as 17+3

23 cannot be expressed as sum of two primes

**For example:**

| Test | Result |
|---|---|
| `print(checkPrimeSum(20))` | yes |
| `print(checkPrimeSum(23))` | no |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def checkPrimeSum(n):
    if n<2:
        return False
    for i in range(2,int(n**0.5)+1):
        if n%i==0:
            return False
    return True
n=int(input())
def checkPrime(n):
    if (n==2):
        return "yes"
    if checkPrime(n) and checkPrime(n-2):
        return "yes"
    return checkPrimeSum(n,n-1)
```

| | Test | Expected | Got |
|---|---|---|---|
| ✓ | `print(checkPrimeSum(20))` | yes | yes ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

**Question 5** Correct Mark 1.00 out of 1.00 Flag question

Complete the recursive function to return Binary Equivalent of an Integer using Recursion.

Sample Test Cases
Test Case 1
Input
10
Output
1010
Test Case 2
Input
257
Output
100000001

**For example:**

| Test | Result |
|---|---|
| `print(binaryNumber(10))` | 1010 |
| `print(binaryNumber(257))` | 100000001 |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def binaryNumber(n):
    return format(n,'b')
```

| | Test | Expected | Got |
|---|---|---|---|
| ✓ | `print(binaryNumber(10))` | 1010 | 1010 ✓ |
| ✓ | `print(binaryNumber(257))` | 100000001 | 100000001 ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Finish review

# CD23321-Python Programming for Design-2025 Batch

| | |
|---|---|
| Started on | Tuesday, 1 September 2026, 9:36 PM |
| State | Finished |
| Completed on | Tuesday, 1 September 2026, 10:50 PM |
| Time taken | 1 hour 13 mins |
| Marks | 10.00/10.00 |
| Grade | **100.00** out of 100.00 |

Quiz navigation

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Finish review

## Question 1
Correct Mark 1.00 out of 1.00 Flag question

A pangram is a sentence where every letter of the English alphabet appears at least once.

Given a string sentence containing only lowercase English letters, return true if sentence is a pangram, or false otherwise.

Example 1:

Input:

thequickbrownfoxjumpsoverthelazydog

Output:

true

Explanation: sentence contains at least one of every letter of the English alphabet.

Example 2:

Input:

arvijayakumar

Output: false

Constraints:

1 <= sentence.length <= 1000

sentence consists of lowercase English letters.

**For example:**

| Test | Result |
|---|---|
| print(checkPangram('thequickbrownfoxjumpsoverthelazydog')) | true |
| print(checkPangram('arvijayakumar')) | false |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def checkPangram(s):
    l=set(s)
    if len (l)==26:
        return "true"
    else:
        return "false"
```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | print(checkPangram('thequickbrownfoxjumpsoverthelazydog')) | true | true | ✓ |
| ✓ | print(checkPangram('arvijayakumar')) | false | false | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 2
Correct Mark 1.00 out of 1.00 Flag question

Consider the below words as key words and check the given input is key word or not.

keywords: {break, case, continue, default, defer, else, for, func, goto, if, map, range, return, struct, type, var}

Input format:

Take string as an input from stdin.

Output format:

Print the word is key word or not.

Example Input:

break

Output:

break is a keyword

Example Input:

IF

Output:

IF is not a keyword

**For example:**

| Input | Result |
|---|---|
| break | break is a keyword |
| IF | IF is not a keyword |

**Answer:** (penalty regime: 0 %)

```
import keyword
s=input()
if keyword.iskeyword(s) or s in keyword.softkwlist:
    print(s+" is a keyword")
else:
    print(s+" is not a keyword")
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | break | break is a keyword | break is a keyword | ✓ |
| ✓ | IF | IF is not a keyword | IF is not a keyword | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 3
Correct Mark 1.00 out of 1.00 Flag question

Given a string, determine if it is a palindrome, considering only alphanumeric characters and ignoring cases.

**Note:** For the purpose of this problem, we define empty string as valid palindrome.

**Example 1:**

```
Input:
A man, a plan, a canal: Panama

Output:
1
```

**Example 2:**

```
Input:
race a car

Output:
0
```

**Constraints:**

- `s` consists only of printable ASCII characters.

**Answer:** (penalty regime: 0 %)

```
s=input()
s="".join(c.lower()for c in s if c.isalnum())
if s==s[::-1]:
    print(1)
else:
    print(0)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | A man, a plan, a canal: Panama | 1 | 1 | ✓ |
| ✓ | race a car | 0 | 0 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 4
Correct Mark 1.00 out of 1.00 Flag question

Find if a String2 is substring of String1. If it is, return the index of the first occurrence. else return -1.

**Sample Input 1**

thistest123string

123

**Sample Output 1**

8

**Answer:** (penalty regime: 0 %)

```
s1=input()
s2=input()

ind=s1.find(s2)
print(ind)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | thistest123string<br>123 | 8 | 8 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 5
Correct Mark 1.00 out of 1.00 Flag question

Write a Python program to get one string and reverses a string. The input string is given as an array of characters `char[]`.

You may assume all the characters consist of printable ascii characters.

**Example 1:**

```
Input:
hello
Output:
olleh
```

**Example 2:**

```
Input:
Hannah
Output:
hannaH
```

**Answer:** (penalty regime: 0 %)

```
char=input()
rev=char[::-1]
print(rev)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | hello | olleh | olleh | ✓ |
| ✓ | Hannah | hannaH | hannaH | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 6
Correct Mark 1.00 out of 1.00 Flag question

Given a string s containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid.

An input string is valid if:

Open brackets must be closed by the same type of brackets.

Open brackets must be closed in the correct order.

Constraints:

1 <= s.length <= 10^4

s consists of parentheses only '()[]{}'.

**For example:**

| Test | Result |
|---|---|
| print(ValidParenthesis("()")) | true |
| print(ValidParenthesis("()[]{}")) | true |
| print(ValidParenthesis("(]")) | false |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def ValidParenthesis(s):
    stack=[]
    pair={')':'(',']':'[','}':'{'}

    for ch in s:
        if ch in '([{':
            stack.append(ch)
        else:
            if not stack or stack[-1] !=pair[ch]:
                return "false"
            stack.pop()
    return "true" if not stack else "false"
```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | print(ValidParenthesis("()")) | true | true | ✓ |
| ✓ | print(ValidParenthesis("()[]{}")) | true | true | ✓ |
| ✓ | print(ValidParenthesis("(]")) | false | false | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 7
Correct Mark 1.00 out of 1.00 Flag question

Given a string S which is of the format USERNAME@DOMAIN.EXTENSION, the program must print the EXTENSION, DOMAIN, USERNAME in the reverse order.

**Input Format:**

The first line contains S.

**Output Format:**

The first line contains EXTENSION.
The second line contains DOMAIN.
The third line contains USERNAME.

**Boundary Condition:**

1 <= Length of S <= 100

**Example Input/Output 1:**

Input:

abcd@gmail.com

Output:

com
gmail
abcd

**For example:**

| Input | Result |
|---|---|
| arvijayakumar@rajalakshmi.edu.in | edu.in<br>rajalakshmi<br>arvijayakumar |

**Answer:** (penalty regime: 0 %)

```
s=input()
u,rest=s.split('@')
part=rest.split('.')
domain=part[0]
ex='.'.join(part[1:])
print(ex)
print(domain)
print(u)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | abcd@gmail.com | com<br>gmail<br>abcd | com<br>gmail<br>abcd | ✓ |
| ✓ | arvijayakumar@rajalakshmi.edu.in | edu.in<br>rajalakshmi<br>arvijayakumar | edu.in<br>rajalakshmi<br>arvijayakumar | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 8
Correct Mark 1.00 out of 1.00 Flag question

Given a **non-empty** string s and an abbreviation abbr, return whether the string matches with the given abbreviation.

A string such as "word" contains only the following valid abbreviations:

["word", "1ord", "w1rd", "wo1d", "wor1", "2rd", "w2d", "wo2", "1o1d", "1or1", "w1r1", "1o2", "2r1", "3d", "w3", "4"]

Notice that only the above abbreviations are valid abbreviations of the string "word". Any other string is not a valid abbreviation of "word".

**Note:**

Assume s contains only lowercase letters and abbr contains only lowercase letters and digits.

**Example 1:**

**Input**

internationalization

i12iz4n

**Output**

true

**Explanation**

Given **s** = "internationalization", **abbr** = "i12iz4n":

Return true.

**Example 2:**

**Input**

apple

a2e

**Output**

false

**Explanation**

Given **s** = "apple", **abbr** = "a2e":

Return false.

**Answer:** (penalty regime: 0 %)

```
s=input()
abb=input()
i=0;
j=0

while i<len(s) and j<len(abb):
    if abb[j].isalpha():
        if s[i]!=abb[j]:
            print("false")
            break
        i+=1
        j+=1
    else:
        if abb[j]=='0':
            print("false")
            break

        num=0
        while j<len(abb) and abb[j].isdigit():
            num=num*10+int(abb[j])
            j+=1
        i+=num
else:
    if i==len(s) and j==len(abb):
        print("true")
    else:
        print("false")
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | internationalization<br>i12iz4n | true | true | ✓ |
| ✓ | apple<br>a2e | false | false | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 9
Correct Mark 1.00 out of 1.00 Flag question

Assume that the given string has enough memory.

Don't use any extra space(IN-PLACE)

**Sample Input 1**

a2b4c6

**Sample Output 1**

aabbbbcccccc

**Answer:** (penalty regime: 0 %)

```
s=input()
result=""
i=0
while i<len(s):
    ch=s[i]
    i+=1
    num=""
    while i<len(s) and s[i].isdigit():
        num+=s[i]
        i+=1
    result+=ch*int(num)
print(result)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | a2b4c6 | aabbbbcccccc | aabbbbcccccc | ✓ |
| ✓ | a12b3d4 | aaaaaaaaaaaabbbdddd | aaaaaaaaaaaabbbdddd | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

## Question 10
Correct Mark 1.00 out of 1.00 Flag question

The program must accept **N** series of keystrokes as string values as the input. The character **^** represents undo action to clear the last entered keystroke. The program must print the string typed after applying the undo operations as the output. If there are no characters in the string then print **-1** as the output.

**Boundary Condition(s):**

1 <= N <= 100
1 <= Length of each string <= 100

**Input Format:**

The first line contains the integer N.
The next N lines contain a string on each line.

**Output Format:**

The first N lines contain the string after applying the undo operations.

**Example Input/Output 1:**

Input:

3
Hey ^ goooo^^glee^
lucke^y ^charr^ms
ora^^nge^^^^

Output:

Hey google
luckycharms
-1

**Answer:** (penalty regime: 0 %)

```
n=int(input())

for _ in range (n):
    s=input()
    stack=[]

    for c in s:
        if c=='^':
            if stack:
                stack.pop()
        else:
            stack.append(c)
    if stack:
        print(''.join(stack))
    else:
        print(-1)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 3<br>Hey ^ goooo^^glee^<br>lucke^y ^charr^ms<br>ora^^nge^^^^ | Hey google<br>luckycharms<br>-1 | Hey google<br>luckycharms<br>-1 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Finish review

# CD23321 Python Programming for Design-2025 Batch

| | |
|---|---|
| Assignment | Assignment 4 Python Programming for Design-2025 Batch |
| Date | 19-12-2025 |
| Completed by | Priya Dharshini, Batch-2025, 111778 |
| Time Limit | 3 hour 0 minutes |
| Points | 100.00 |
| Grade | 100.00 out of 100.00 |

Question 1 Correct Mark 100.00 out of 100.00 Flag question

You are given an integer `num`. Consider a binary number `x` that can be written in powers of two where `x` is the sum of some of the powers of two (possibly all or none).

Return the number of binary numbers `x` such that the binary representation of `x` contains no more than `num` consecutive `1`'s.

For example, if `num = 2`, we cannot have `x = 3` or `x = 7` because their binary representations `"11"` and `"111"` have more than `2` consecutive `1`'s.

Since the answer may be too large, return it modulo `10^9 + 7`.

**Constraints**

- `1 <= num <= 10^5`

Input format:

Input consists of a single integer `num`.

Output format:

Print the number of binary numbers `x` such that the binary representation of `x` contains no more than `num` consecutive `1`'s, modulo `10^9 + 7`.

Sample Input 1:

```
3
```

Sample Output 1:

```
5
```

Explanation:

The valid numbers are `1, 2, 4, 5, 6`. The binary representations are `"1"`, `"10"`, `"100"`, `"101"`, and `"110"`. Each has at most `3` consecutive `1`'s.

Sample Input 2:

```
1
```

Sample Output 2:

```
2
```

Explanation:

The valid numbers are `1` and `2`. The binary representations are `"1"` and `"10"`.

Sample Input 3:

```
2
```

Sample Output 3:

```
4
```

Explanation:

The valid numbers are `1, 2, 4, 5`. The binary representations are `"1"`, `"10"`, `"100"`, and `"101"`.

**For example:**

| Input | Result |
|---|---|
| `3` | `5` |

Answer: (penalty regime: 0 %)

```
Reset answer
```

Question 2 Correct Mark 100.00 out of 100.00 Flag question

Write a program to determine the common elements in the given 2 arrays and print only the non-repeating elements in ascending order.

Input format:

The first line contains space-separated values denoting the size of the two arrays in range 5 and more respectively.

The next two lines contain space-separated integer values of the two arrays.

Output format:

Print the common non-repeating elements in ascending order.

Sample Input 1:

```
5 6
1 2 3 4 5
3 4 5 6 7 8
```

Sample Output 1:

```
3 4 5
```

Sample Input 2:

```
5 5
1 2 3 4 5
3 4 5 6 7
```

Sample Output 2:

```
3 4 5
```

Sample Input 3:

```
5 5
1 2 3 4 5
6 7 8 9 10
```

Sample Output 3:

```
No common elements
```

**For example:**

| Input | Result |
|---|---|
| `5 6`<br>`1 2 3 4 5`<br>`3 4 5 6 7 8` | `3 4 5` |
| `5 5`<br>`1 2 3 4 5`<br>`6 7 8 9 10` | `No common elements` |

Answer: (penalty regime: 0 %)

```
Reset answer
```

Question 3 Correct Mark 100.00 out of 100.00 Flag question

Given a table and a column integer N, the task is to find the count of distinct values in the table given column is equal to N.

**Example:**

Input : N = 2

Table = {{1, 1, 1},

{1, 2, 1},

{1, 3, 1},

{1, 4, 1}}

Output : 4

Explanation: In 2nd column we have 1, 2, 3 and 4 i.e. 4 distinct values.

Input : N = 3

Table = {{1, 2, 3},

{4, 5, 6},

{7, 8, 9},

{10, 11, 12}}

Output : 4

Explanation: In 3rd column we have 3, 6, 9 and 12 i.e. 4 distinct values.

Therefore, the required output is 4.

**For example:**

| Input | Result |
|---|---|
| `4 3`<br>`1 1 1`<br>`1 2 1`<br>`1 3 1`<br>`1 4 1` | `4` |

Answer: (penalty regime: 0 %)

```
Reset answer
```

Question 4 Correct Mark 100.00 out of 100.00 Flag question

Given a string S and integers L and R, consisting of a unique substring in the range L to R. A substring is a sequence of characters within a string that is entirely contained within another string. Note that the substring is unique.

**Example:**

Input: S = "abcdefghijk", L = 3, R = 5

Output: "cde"

Explanation: The substring in the range 3 to 5 is "cde". This is the only substring of length 3 in this range.

Input: S = "abcdefghijk", L = 1, R = 3

Output: "abc"

Explanation: The substring in the range 1 to 3 is "abc". This is the only substring of length 3 in this range.

**For example:**

| Input | Result |
|---|---|
| `abcdefghijk`<br>`3`<br>`5` | `cde` |

Answer: (penalty regime: 0 %)

```
Reset answer
```

Question 5 Correct Mark 100.00 out of 100.00 Flag question

There is a self-training program which runs every day for one month. We either log in or do the balance(not both) anytime.

There is one log file and one balance log file which records all login / balance transactions.

Record data in integer id and time in hours of a day.

Output:

Print the total minutes in hours of login for each id, in ascending order of id.

Example Input:

```
5
1 10
2 6
1 20
2 16
3 18
5
1 15
2 20
3 21
3 2
1 11
```

Example Output:

```
1 14
2 24
3 3
```

**For example:**

| Input | Result |
|---|---|
| `5`<br>`1 10`<br>`2 6`<br>`1 20`<br>`2 16`<br>`3 18`<br>`5`<br>`1 15`<br>`2 20`<br>`3 21`<br>`3 2`<br>`1 11` | `1 14`<br>`2 24`<br>`3 3` |

Answer: (penalty regime: 0 %)

```
Reset answer
```

Question 6 Correct Mark 100.00 out of 100.00 Flag question

There is a self-training program which runs every day for one month. We either login or do the balance(not both) anytime.

There is one log file and one balance log file which records all login / balance transactions.

Record data in integer id and time in hours of a day.

Output:

Print the total minutes in hours of login for each id, in ascending order of id.

Example Input:

```
5
1 10
2 6
1 20
2 16
3 18
5
1 15
2 20
3 21
3 2
1 11
```

Example Output:

```
1 14
2 24
3 3
```

**For example:**

| Input | Result |
|---|---|
| `5`<br>`1 10`<br>`2 6`<br>`1 20`<br>`2 16`<br>`3 18`<br>`5`<br>`1 15`<br>`2 20`<br>`3 21`<br>`3 2`<br>`1 11` | `1 14`<br>`2 24`<br>`3 3` |

Answer: (penalty regime: 0 %)

```
Reset answer
```

Question 7 Correct Mark 100.00 out of 100.00 Flag question

Given a string of integers `s`, a repeating subsequence is a subsequence that occurs more than once in the string.

Return the longest repeating subsequence. If there is more than one answer, return the one that comes first in lexicographical order. If there is no repeating subsequence, return an empty string.

**Example:**

Input: `s = "123123"`

Output: `"123"`

Explanation: The subsequences `"123"`, `"12"`, `"23"`, `"13"`, `"1"`, `"2"`, and `"3"` all occur more than once. Since `"123"` is the longest, it is the answer.

Input: `s = "1234567"`

Output: `""`

Explanation: There are no repeating subsequences, so return an empty string.

**For example:**

| Input | Result |
|---|---|
| `123123` | `123` |

Answer: (penalty regime: 0 %)

```
Reset answer
```

Question 8 Correct Mark 100.00 out of 100.00 Flag question

Check if a set is a subset of another set.

**For example:**

| Input | Result |
|---|---|
| `orange`<br>`mango`<br>`grapes`<br>`orange`<br>`apple`<br>`mango`<br>`grapes`<br>`pineapple` | `orange is not present in set2` |

Answer: (penalty regime: 0 %)

```
Reset answer
```

Question 9 Correct Mark 100.00 out of 100.00 Flag question

Check if a set is a subset of another set.

**For example:**

| Input | Result |
|---|---|
| `orange`<br>`mango`<br>`grapes`<br>`orange`<br>`apple`<br>`mango`<br>`grapes`<br>`pineapple` | `orange is not present in set2` |

Answer: (penalty regime: 0 %)

```
Reset answer
```

Question 10 Correct Mark 100.00 out of 100.00 Flag question

Check if a set is a subset of another set.

**For example:**

| Input | Result |
|---|---|
| `orange`<br>`mango`<br>`grapes`<br>`orange`<br>`apple`<br>`mango`<br>`grapes`<br>`pineapple` | `orange is not present in set2` |

Answer: (penalty regime: 0 %)

```
Reset answer
```

Finish review

Quiz navigation

1 2 3 4 5 6 7 8 9 10

Finish attempt ...

Time left 2:59:51

# CD23321-Python Programming for Design-2025 Batch

| | |
|---|---|
| **Started on** | Wednesday, 7 October 2026, 11:12 PM |
| **State** | Finished |
| **Completed on** | Wednesday, 7 October 2026, 11:20 PM |
| **Time taken** | 7 mins 53 secs |
| **Marks** | 5.00/5.00 |
| **Grade** | **100.00** out of 100.00 |

Quiz navigation

1 2 3 4 5

Show one page at a time

Finish review

**Question 1** Correct Mark 1.00 out of 1.00 Flag question

A sentence is a list of words that are separated by a single space with no leading or trailing spaces. Each word consists of lowercase and uppercase English letters.

A sentence can be shuffled by appending the 1-indexed word position to each word then rearranging the words in the sentence.

For example, the sentence "This is a sentence" can be shuffled as "sentence4 a2 This1 is3" or "a2 sentence4 This1 is3".

Given a shuffled sentence s containing no more than 9 words, reconstruct and return the original sentence.

Example 1:

**Input:**
s2 sentence4 This1 a3

**Output:**
This is a sentence

Explanation: Sort the words in s to their original positions "This1 is2 a3 sentence4", then remove the numbers.

Example 2:

**Input:**
Myself2 Me1 I4 and3

**Output:**
Me Myself and I

Explanation: Sort the words in s to their original positions "Me1 Myself2 and3 I4", then remove the numbers.

Constraints:

2 <= s.length <= 200

s consists of lowercase and uppercase English letters, spaces, and digits from 1 to 9.

The number of words in s is between 1 and 9.

The words in s are separated by a single space.

s contains no leading or trailing spaces.

**Answer:** (penalty regime: 0 %)

```
s = input().split()
ans = [""] * len(s)
for word in s:
    pos = int(word[-1])
    ans[pos - 1] = word[:-1]
print(" ".join(ans))
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | is2 sentence4 This1 a3 is2 a sentence4 This1 a3 | This is a sentence | This is a sentence | ✓ |

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 2** Correct Mark 1.00 out of 1.00 Flag question

There is a malfunctioning keyboard where some letter keys do not work. All other keys on the keyboard work properly.

Given a string text of words separated by a single space (no leading or trailing spaces) and a string brokenLetters of all distinct letter keys that are broken, return the number of words in text you can fully type using this keyboard.

Example 1:

Input: text = "hello world", brokenLetters = "ad"

Output: 1

Explanation: We cannot type "world" because the 'd' key is broken.

**For example:**

| Input | Result |
|---|---|
| hello world<br>ad | 1 |
| Faculty Upskilling in Python Programming<br>ab | 2 |

**Answer:** (penalty regime: 0 %)

```
text = input().split()
broken = input().lower()

count = 0

for word in text:
    ok = True

    for ch in word.lower():
        if ch in broken:
            ok = False
            break

    if ok:
        count += 1

print(count)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | hello world<br>ad | 1 | 1 | ✓ |
| ✓ | Welcome to REC<br>o | 1 | 1 | ✓ |
| ✓ | Faculty Upskilling in Python Programming<br>ab | 2 | 2 | ✓ |

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 3** Correct Mark 1.00 out of 1.00 Flag question

Given a number, convexent it into corresponding alphabet.

| Input | Output |
|---|---|
| 1 | A |
| 26 | Z |
| 27 | AA |
| 676 | YZ |

**Input Format**

Input is an integer

**Output Format**

Print the alphabets

**Constraints**

1 <= num <= 4294967295

**Sample Input 1**

26

**Sample Output 1**

Z

**For example:**

| Test | Result |
|---|---|
| print(excelNumber(26)) | Z |

Reset answer

**Answer:** (penalty regime: 0 %)

```
def excelNumber(n):
    ans = ""

    while n > 0:
        n = n - 1
        ans = chr(65 + n) + ans
        n = n // 26

    return ans
```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | print(excelNumber(26)) | Z | Z | ✓ |
| ✓ | print(excelNumber(27)) | AA | AA | ✓ |

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 4** Correct Mark 1.00 out of 1.00 Flag question

Given an array of strings *words*, *return the words that can be typed using letters of the alphabet on only one row of American keyboard like the image below.*

In the **American keyboard**:

- the first row consists of the characters *"qwertyuiop"*,
- the second row consists of the characters *"asdfghjkl"*, and
- the third row consists of the characters *"zxcvbnm"*.




**Example 1:**

Input: words = ["Hello","Alaska","Dad","Peace"]

Output: ["Alaska","Dad"]

**Example 2:**

Input: words = ["omk"]

Output: []

**Example 3:**

Input: words = ["adsdf","sfd"]

Output: ["adsdf","sfd"]

**For example:**

| Input | Result |
|---|---|
| 4<br>Hello<br>Alaska<br>Dad<br>Peace | Alaska<br>Dad |
| 2<br>omk<br>adsfd<br>sfd | adsfd<br>sfd |

**Answer:** (penalty regime: 0 %)

```
n = int(input())

row1 = "qwertyuiop"
row2 = "asdfghjkl"
row3 = "zxcvbnm"

found = False

for i in range(n):
    word = input()
    w = word.lower()

    if all(c in row1 for c in w) or all(c in row2 for c in w) or all(c in row3 for c in w):
        print(word)
        found = True

if not found == False:
    print("No words")
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 4<br>Hello<br>Alaska<br>Dad<br>Peace | Alaska<br>Dad | Alaska<br>Dad | ✓ |
| ✓ | 1<br>omk | No words | No words | ✓ |
| ✓ | 2<br>adsfd<br>sfd | adsfd<br>sfd | adsfd<br>sfd | ✓ |

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

**Question 5** Correct Mark 1.00 out of 1.00 Flag question

The program must accept a string S which represents a series of keystrokes as the input. There are two special keystrokes denoted by "/z" for undo operation and "/y" for redo operation. The undo operation removes the last typed character and redo operations reverses the immediately done last undo operations. The program must accept the keystrokes and perform all the operations.Lastly and finally, the program must print the string after applying all the operations as the output.

**Boundary Condition(s):**

1 <= Length of string <= 1000

**Input Format:**

The first line contains the string S.

**Output Format:**

The first line contains the modified string.

**Example Input/Output 1:**

Input:

missef/z/z/y/ypdf/z/z/y/y.

Output:

missepdf

Explanation:

The characters **missef** are followed by three undo operations which deletes last three characters.

So the string **missef** becomes **miss**.

The three undo operations are followed by one redo operation which reverses the last undo operation.

So the string **miss** becomes **missa**.

Then the characters pdf are entered which are followed by two undo operations so the last two characters are removed.

So the string becomes **missppa**.

Then a redo operation is applied which reverses the last undo operation and so the string **missppa** becomes **missepdf**.

The characters are followed by **y** so the final string is missepdf.

**Answer:** (penalty regime: 0 %)

```
S = input()

a = []
redo = []

i = 0

while i < len(S):
    if s[i:i+2] == "/z":
        if len(a) > 0:
            redo.append(a.pop())
        i += 2
    elif s[i:i+2] == "/y":
        if len(redo) > 0:
            a.append(redo.pop())
        i += 2
    else:
        a.append(S[i])
        redo = []
        i += 1

print("".join(a))
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | missef/z/z/y/ypdf/z/z/y/y | missepdf | missepdf | ✓ |

Passed all tests! ✓

Marks for this submission: 1.00/1.00.

Finish review

# CS2C221-Python Programming for Design-2025 Batch



Question 1 Correct Mark 100.00 out of 100.00 Flag question

Take the first letter from the first name and the second letter of second name.

Input: Ram Kumar
Output: RA (Take the first letter of Ram and second letter of Kumar)

Input: Jack Ryan
Output: JY (Take the first letter of Jack and second letter of Ryan)

Input: Ram Ram
Output: RR (Take the first letter of Ram and second letter of Ram)

Constraints: Consider only one word between

Sample Input

```
Ram
Kumar
```

Sample Output

```
RA
```

For example:

| Input | Result |
|---|---|
| Ram<br>Kumar | RA |

Answer: (penalty regime: 0 %)

```
# Read the first name and last name
first_name = input()
last_name = input()

# Extract the first letter of the first name and the second letter of the last name
result = first_name[0] + last_name[1]

# Print the result in uppercase
print(result.upper())
```

| Input | Expected | Got |
|---|---|---|
| Ram<br>Kumar | RA | RA |
| Jack<br>Ryan | JY | JY |
| Ram<br>Ram | RR | RR |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 2 Correct Mark 100.00 out of 100.00 Flag question

Create a program that will take a string and replace every character in the string with the character standing in front of it and replace the first character with the last character to complete the cycle. This can be done for any number of times. Take the number of times the string has to be encrypted as input.

Example: If the string is "apple" and the number of times is 1, then the output should be "papel".

1. The last character moves to the front.
2. All other characters are shifted one position to the right.

Sample Input 1

```
apple
1
```

Sample Output 1

```
papel
```

Sample Input 2

```
hello
2
```

Sample Output 2

```
lohel
```

For example:

| Input | Result |
|---|---|
| apple<br>1 | papel |
| hello<br>2 | lohel |

Answer: (penalty regime: 0 %)

```
# Read the input string and the number of times
s = input()
n = int(input())

# Perform the encryption n times
for _ in range(n):
    s = s[-1] + s[:-1]

# Print the final encrypted string
print(s)
```

| Input | Expected | Got |
|---|---|---|
| apple<br>1 | papel | papel |
| hello<br>2 | lohel | lohel |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 3 Correct Mark 100.00 out of 100.00 Flag question

Given a positive number check for a prime number.

Example:

Input: 7

Output: Prime

Input: 10

Output: Not Prime

Input: 1

Output: Not Prime

Input: -5

Output: Not Prime

Constraints: Consider only positive numbers

Sample Input 1

```
7
```

Sample Output 1

```
Prime
```

Sample Input 2

```
10
```

Sample Output 2

```
Not Prime
```

For example:

| Input | Result |
|---|---|
| 7 | Prime |
| 10 | Not Prime |

Answer: (penalty regime: 0 %)

```
# Read the input number
num = int(input())

# Check if the number is prime
if num <= 1:
    print("Not Prime")
else:
    for i in range(2, num):
        if num % i == 0:
            print("Not Prime")
            break
    else:
        print("Prime")
```

| Input | Expected | Got |
|---|---|---|
| 7 | Prime | Prime |
| 10 | Not Prime | Not Prime |
| 1 | Not Prime | Not Prime |
| -5 | Not Prime | Not Prime |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 4 Correct Mark 100.00 out of 100.00 Flag question

Given a number, convert it into its corresponding alphabet.

Input: 1

Output: A

Input: 26

Output: Z

Input: 27

Output: Not Possible

Input: 0

Output: Not Possible

Input: -5

Output: Not Possible

Constraints: Consider only positive numbers

Sample Input 1

```
1
```

Sample Output 1

```
A
```

Sample Input 2

```
26
```

Sample Output 2

```
Z
```

For example:

| Input | Result |
|---|---|
| 1 | A |
| 26 | Z |

Answer: (penalty regime: 0 %)

```
# Read the input number
num = int(input())

# Check if the number is within the range of 1 to 26
if 1 <= num <= 26:
    # Convert the number to its corresponding alphabet
    print(chr(num + 64))
else:
    print("Not Possible")
```

| Input | Expected | Got |
|---|---|---|
| 1 | A | A |
| 26 | Z | Z |
| 27 | Not Possible | Not Possible |
| 0 | Not Possible | Not Possible |
| -5 | Not Possible | Not Possible |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 5 Correct Mark 100.00 out of 100.00 Flag question

Write a program to find the sum of the first n terms of the series 1 + 11 + 111 + 1111 + 11111 + ...

Test Case 1

Input: 5

Output: 12345

Test Case 2

Input: 8

Output: 13578

Test Case 3

Input: 4

Output: 1234

Test Case 4

Input: 2

Output: 12

Constraint: Consider only positive integers

Constraint: Consider that the sum is always positive and will not exceed the integer limit of the system.

Sample Input 1

```
5
```

Sample Output 1

```
12345
```

Sample Input 2

```
8
```

Sample Output 2

```
13578
```

For example:

| Test | Result |
|---|---|
| print(sum_of_series(5)) | 12345 |

Answer: (penalty regime: 0 %)

```
def sum_of_series(n):
    total = 0
    term = 1
    for i in range(n):
        total += term
        term = term * 10 + 1
    return total

print(sum_of_series(5))
```

| Test | Expected | Got |
|---|---|---|
| print(sum_of_series(5)) | 12345 | 12345 |
| print(sum_of_series(8)) | 13578 | 13578 |
| print(sum_of_series(4)) | 1234 | 1234 |
| print(sum_of_series(2)) | 12 | 12 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 6 Correct Mark 100.00 out of 100.00 Flag question

Write a program to find the sum of the series 1 + 11 + 111 + 1111 + 11111 + ...

The first line contains an integer n.

The output is a single integer which is the sum of the first n terms of the series.

Test Case 1

Input: 5

Output: 12345

Test Case 2

Input: 8

Output: 13578

Test Case 3

Input: 4

Output: 1234

Test Case 4

Input: 2

Output: 12

Constraint: Consider only positive integers

Constraint: Consider that the sum is always positive and will not exceed the integer limit of the system.

Sample Input 1

```
5
```

Sample Output 1

```
12345
```

Sample Input 2

```
8
```

Sample Output 2

```
13578
```

For example:

| Test | Result |
|---|---|
| print(sum_of_series(5)) | 12345 |

Answer: (penalty regime: 0 %)

```
def sum_of_series(n):
    total = 0
    term = 1
    for i in range(n):
        total += term
        term = term * 10 + 1
    return total

print(sum_of_series(5))
```

| Test | Expected | Got |
|---|---|---|
| print(sum_of_series(5)) | 12345 | 12345 |
| print(sum_of_series(8)) | 13578 | 13578 |
| print(sum_of_series(4)) | 1234 | 1234 |
| print(sum_of_series(2)) | 12 | 12 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 7 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 8 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 9 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 10 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 11 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 12 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 13 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 14 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 15 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 16 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 17 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 18 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 19 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 20 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 21 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 22 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 23 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 24 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 25 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 26 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 27 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 28 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 29 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 30 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 31 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 32 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Passed all tests! ✓

Marks for this submission: 100.00.

Question 33 Correct Mark 100.00 out of 100.00 Flag question

Write a program to print the following pattern.

Input: 4

Output:

```
1
2 3
4 5 6
7 8 9 10
```

Input: 5

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

For example:

| Input | Result |
|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6<br>7 8 9 10<br>11 12 13 14 15 |

Answer: (penalty regime: 0 %)

```
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

| Input | Expected | Got |
|---|---|---|
| 4 | 1<br>2 3<br>4 5 6<br>7 8 9 10 | 1<br>2 3<br>4 5 6<br>7 8 9 10 |
| 5 | 1<br>2 3<br>4 5 6

# CD23321-Python Programming for Design-2025 Batch

Started on Thursday, 13 November 2025, 10:52 PM

State Finished

Completed on Thursday, 13 November 2025, 10:54 PM

Time taken 1 min 19 secs

Marks 10.00/10.00

Grade 100.00 out of 100.00

Question 1 Correct Mark 2.00 out of 2.00

Create a Python program to delete a specific line from a text file based on a given line number.

Description:

- Input:
  - A text file with multiple lines.
  - An integer line_number.
- Output:
  - The content of the text file after the specified line number is removed (the "output.txt" file).

Example:

- Input File Contents:
  Time tells
  Use time
  Use time
  Use Now!
- Line_number = 2
- Output:
  Time tells
  Use Now!

Constraints:

Line_number

Input File Name

Output File Name

Test Cases:

| Test | Input | Result |
|---|---|---|
| with open("output.txt", "r") as file: print(file.read()) | input.txt 2 | Time tells<br>Use Now! |
| with open("output.txt", "r") as file: print(file.read()) | input.txt 1 | Use time<br>Use Now! |

Answer: (penalty regime: 0 %)

```
line_number = int(input())
with open("input.txt", "r") as file:
    lines = file.readlines()
if 1 <= line_number <= len(lines):
    del lines[line_number - 1]
with open("output.txt", "w") as file:
    file.writelines(lines)
```

| Test | Input | Expected | Got |
|---|---|---|---|
| with open("output.txt", "r") as file: print(file.read()) | 2<br>input.txt | Time tells<br>Use Now! | Time tells<br>Use Now! |
| with open("output.txt", "r") as file: print(file.read()) | 1<br>input.txt | Use time<br>Use Now! | Use time<br>Use Now! |

Passed all tests! ✓

Marks for this submission: 2.00/2.00.

Question 2 Correct Mark 2.00 out of 2.00

Write a Python program to remove the contents of a specific line in a text file based on a given line number.

Description:

- Input:
  - A text file with multiple lines.
  - A line number to remove.
- Output:
  - The updated file with the specified line's contents removed in the "output.txt".

Example:

- Input File Contents:
  Time tells
  Use time
  Use Now!
- Line_number = 2
- Output:
  Time tells
  Use Now!

Constraints:

Line_number

Input File Name

Output File Name

Test Cases:

| Test | Input | Result |
|---|---|---|
| with open("output.txt", "r") as file: print(file.read()) | input.txt 2 | Time tells<br>Use Now! |
| with open("output.txt", "r") as file: print(file.read()) | input.txt 1 | Use time<br>Use Now! |

Answer: (penalty regime: 0 %)

```
line_number = int(input())
with open("input.txt", "r") as file:
    lines = file.readlines()
if 1 <= line_number <= len(lines):
    del lines[line_number - 1]
with open("output.txt", "w") as file:
    file.writelines(lines)
```

| Test | Input | Expected | Got |
|---|---|---|---|
| with open("output.txt", "r") as file: print(file.read()) | input.txt 2 | Time tells<br>Use Now! | Time tells<br>Use Now! |
| with open("output.txt", "r") as file: print(file.read()) | input.txt 1 | Use time<br>Use Now! | Use time<br>Use Now! |

Passed all tests! ✓

Marks for this submission: 2.00/2.00.

Question 3 Correct Mark 2.00 out of 2.00

Write a Python program to count the frequency of each word in a given text file.

Description:

- Input:
  - A text file.
  - String to Count
- Output:
  - A list of words with their corresponding frequency count to be wrote in the "output.txt"

Example:

- Input File Content:
  apple orange apple banana apple orange
- Output:
  apple: 3
  orange: 2
  banana: 1

Test Cases:

| Test | Input | Result |
|---|---|---|
| with open("output.txt", "r") as file: print(file.read()) | input.txt | apple: 3<br>orange: 2<br>banana: 1 |

Answer: (penalty regime: 0 %)

```
from collections import Counter
with open("input.txt", "r") as file:
    words = file.read().lower().split()
word_counts = Counter(words)
with open("output.txt", "w") as file:
    for word, count in word_counts.items():
        file.write(f"{word}: {count}\n")
```

| Test | Input | Expected | Got |
|---|---|---|---|
| with open("output.txt", "r") as file: print(file.read()) | input.txt | apple: 3<br>orange: 2<br>banana: 1 | apple: 3<br>orange: 2<br>banana: 1 |
| with open("output.txt", "r") as file: print(file.read()) | input.txt | hello: 2<br>world: 2<br>programming: 1<br>is: 1<br>fun: 1<br>welcome: 1<br>to: 1<br>learning: 1<br>python: 1 | hello: 2<br>world: 2<br>programming: 1<br>is: 1<br>fun: 1<br>welcome: 1<br>to: 1<br>learning: 1<br>python: 1 |
| with open("output.txt", "r") as file: print(file.read()) | input.txt | test: 4<br>file: 3<br>this: 2<br>is: 2<br>a: 2<br>line: 2<br>with: 1<br>some: 1<br>words: 1 | test: 4<br>file: 3<br>this: 2<br>is: 2<br>a: 2<br>line: 2<br>with: 1<br>some: 1<br>words: 1 |

Passed all tests! ✓

Marks for this submission: 2.00/2.00.

Question 4 Correct Mark 2.00 out of 2.00

Write a Python program to append a new line to a specific position in a text file, shifting existing lines down.

Description:

- Input:
  - A text file with multiple lines.
  - A line number to insert the new line at.
  - The new line to be inserted.
- Output:
  - The updated file with the new line inserted at the specified position, shifting the existing lines down (in the "output.txt").

Example:

- Input File Contents:
  Time tells
  Use time
  Use Now!
- Line_number = 2
- New_line = "Use time"
- Output:
  Time tells
  Use time
  Use time
  Use Now!

Test Cases:

| Test | Input | Result |
|---|---|---|
| with open("output.txt", "r") as file: print(file.read()) | input.txt 2 Use time | Time tells<br>Use time<br>Use time<br>Use Now! |
| with open("output.txt", "r") as file: print(file.read()) | input.txt 1 Use Now! | Use Now!<br>Time tells<br>Use time<br>Use Now! |

Answer: (penalty regime: 0 %)

```
line_number = int(input())
new_line = input()
with open("input.txt", "r") as file:
    lines = file.readlines()
if 1 <= line_number <= len(lines) + 1:
    lines.insert(line_number - 1, new_line + "\n")
with open("output.txt", "w") as file:
    file.writelines(lines)
```

| Test | Input | Expected | Got |
|---|---|---|---|
| with open("output.txt", "r") as file: print(file.read()) | input.txt 2 Use time | Time tells<br>Use time<br>Use time<br>Use Now! | Time tells<br>Use time<br>Use time<br>Use Now! |
| with open("output.txt", "r") as file: print(file.read()) | input.txt 1 Use Now! | Use Now!<br>Time tells<br>Use time<br>Use Now! | Use Now!<br>Time tells<br>Use time<br>Use Now! |

Passed all tests! ✓

Marks for this submission: 2.00/2.00.

Question 5 Correct Mark 2.00 out of 2.00

Develop a Python program to find the longest word in a text file and count the total number of words in the file.

Description:

- Input:
  - A text file containing multiple lines of text.
- Output:
  - The longest word in the file.

Test Cases:

| Input | Result |
|---|---|
| input.txt | longest word containing |

Answer: (penalty regime: 0 %)

```
with open("input.txt", "r") as file:
    words = file.read().split()
if words:
    longest = max(words, key=len)
    print("Longest word:", longest)
    print("Total number of words:", len(words))
else:
    print("The file is empty.")
```

| Test | Input | Expected | Got |
|---|---|---|---|
| with open("output.txt", "r") as file: print(file.read()) | input.txt | Longest word: longest<br>Total number of words: 5 | Longest word: longest<br>Total number of words: 5 |
| with open("output.txt", "r") as file: print(file.read()) | input.txt | Longest word: programming<br>Total number of words: 6 | Longest word: programming<br>Total number of words: 6 |

Passed all tests! ✓

Marks for this submission: 2.00/2.00.

Question 6 Correct Mark 2.00 out of 2.00

Develop a Python program to find the longest word in a text file and count the total number of words in the file.

Description:

- Input:
  - A text file containing multiple lines of text.
- Output:
  - The longest word in the file.
  - The total number of words in the file.

Test Cases:

| Input | Result |
|---|---|
| input.txt | Longest word: longest<br>Total number of words: 5 |

Answer: (penalty regime: 0 %)

```
with open("input.txt", "r") as file:
    words = file.read().split()
if words:
    longest = max(words, key=len)
    print("Longest word:", longest)
    print("Total number of words:", len(words))
else:
    print("The file is empty.")
```

| Input | Expected | Got |
|---|---|---|
| input.txt | Longest word: longest<br>Total number of words: 5 | Longest word: longest<br>Total number of words: 5 |
| input.txt | Longest word: programming<br>Total number of words: 6 | Longest word: programming<br>Total number of words: 6 |

Passed all tests! ✓

Marks for this submission: 2.00/2.00.

Question 7 Correct Mark 2.00 out of 2.00

Develop a Python program to find the word with the most number of vowels in a file.

Description:

- Input:
  - A text file containing multiple lines of text.
- Output:
  - The word with the most vowels in the file.
  - The number of vowels in the file.

Test Cases:

| Input | Result |
|---|---|
| input.txt | Word with most vowels: education<br>Number of vowels: 5 |

Answer: (penalty regime: 0 %)

```
with open("input.txt", "r") as file:
    words = file.read().split()
if words:
    vowels = "aeiouAEIOU"
    max_word = max(words, key=lambda word: sum(1 for char in word if char in vowels))
    max_vowels = sum(1 for char in max_word if char in vowels)
    print("Word with most vowels:", max_word)
    print("Number of vowels:", max_vowels)
else:
    print("The file is empty.")
```

| Input | Expected | Got |
|---|---|---|
| input.txt | Word with most vowels: education<br>Number of vowels: 5 | Word with most vowels: education<br>Number of vowels: 5 |
| input.txt | Word with most vowels: beautiful<br>Number of vowels: 5 | Word with most vowels: beautiful<br>Number of vowels: 5 |

Passed all tests! ✓

Marks for this submission: 2.00/2.00.

Question 8 Correct Mark 2.00 out of 2.00

Develop a Python program to copy the contents of one file to another file.

Description:

- Input:
  - The contents of the source file copied to the destination file.

Test Cases:

| Input | Result |
|---|---|
| input.txt output.txt | This is the source file.<br>It contains multiple lines of text.<br>This is another line. |

Answer: (penalty regime: 0 %)

```
with open("input.txt", "r") as source_file:
    content = source_file.read()
with open("output.txt", "w") as destination_file:
    destination_file.write(content)
```

| Test | Input | Expected | Got |
|---|---|---|---|
| with open("output.txt", "r") as file: print(file.read()) | input.txt | This is the source file.<br>It contains multiple lines of text.<br>This is another line. | This is the source file.<br>It contains multiple lines of text.<br>This is another line. |
| with open("output.txt", "r") as file: print(file.read()) | input.txt | Hello, world!<br>This is a test file.<br>It contains some text. | Hello, world!<br>This is a test file.<br>It contains some text. |
| with open("output.txt", "r") as file: print(file.read()) | input.txt | Python is awesome.<br>Let's copy this text to another file. | Python is awesome.<br>Let's copy this text to another file. |

Passed all tests! ✓

Marks for this submission: 2.00/2.00.

Question 9 Correct Mark 2.00 out of 2.00

Develop a Python program to copy the contents of one file to another file.

Description:

- Input:
  - The contents of the source file copied to the destination file.

Test Cases:

| Input | Result |
|---|---|
| input.txt output.txt | This is the source file.<br>It contains multiple lines of text.<br>This is another line. |

Answer: (penalty regime: 0 %)

```
with open("input.txt", "r") as source_file:
    content = source_file.read()
with open("output.txt", "w") as destination_file:
    destination_file.write(content)
```

| Test | Input | Expected | Got |
|---|---|---|---|
| with open("output.txt", "r") as file: print(file.read()) | input.txt | This is the source file.<br>It contains multiple lines of text.<br>This is another line. | This is the source file.<br>It contains multiple lines of text.<br>This is another line. |
| with open("output.txt", "r") as file: print(file.read()) | input.txt | Hello, world!<br>This is a test file.<br>It contains some text. | Hello, world!<br>This is a test file.<br>It contains some text. |
| with open("output.txt", "r") as file: print(file.read()) | input.txt | Python is awesome.<br>Let's copy this text to another file. | Python is awesome.<br>Let's copy this text to another file. |

Passed all tests! ✓

Marks for this submission: 2.00/2.00.

Finish review